

EEBus UC Technical Specification

Power Envelope

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1 Scope of the document

This document describes the Use Case "Power Envelope" (short-name: POEN). Chapter 2 specifies the High-Level Use Case. Chapter 3 details the technical solution for SPINE for this Use Case. Within this document, a top-down approach is used to derive the requirements for the technical solution from the High-Level description.

1.1 References

1.1.1 EEBUS documents

[ProtocolSpecification] EEBus_SPINE_TS_ProtocolSpecification.pdf

[ResourceSpecification] EEBus_SPINE_TS_ResourceSpecification.pdf

[SHIP] EEBus_SHIP_TS_Specification_v1.0.1.pdf

1.1.2 Normative references

[RFC2119] IETF RFC 2119: 1997, Key words for use in RFCs to indicate requirement levels
Please see section 1.3.1 for details.

1.2 Terms and definitions

Actor

An Actor models a role within a Use Case definition (e.g. an energy manager or a heat pump).

CEM

Abbreviation for Customer Energy Manager. The CEM is an energy manager located at the user's home or premises or in a cloud application. The energy manager enables energy-optimized operation of the connected devices by harmonizing energy demand and availability.

DSO

Abbreviation for "Distribution System Operator"

EV

Abbreviation for "Electric Vehicle"

Energy Guard

A logical or physical unit capable of managing the energy demand of connected appliances.

Fallback Curve

The limit curves Fallback Maximum Active Power Production Limit Curve ([POEN-031]) and Fallback Maximum Active Power Production Limit Curve ([POEN-032]) are all called Fallback Curve.

Limit Curve

Any curve of Fallback Curve and Power Limitation Curve.

150 LPC

151 Short name of the Use Case "Limitation of Power Consumption"

152 LPP

153 Short name of the Use Case "Limitation of Power Production"

154 PODF

155 Short name of the Use Case "Power Demand Forecast"

156 POEN

157 Power Envelope (short name of this Use Case)

158 Power Envelope

159 A combination of one to four of the Power Limitation Curves of this Use Case is called a Power
160 Envelope.

161 Power Limitation Curve

162 The limit curves Minimum Active Power Consumption Limit Curve ([POEN-011]), Maximum Active
163 Power Consumption Limit Curve ([POEN-012]), Minimum Active Power Production Limit Curve
164 ([POEN-021]) and Maximum Active Power Production Limit Curve ([POEN-022]) are all called Power
165 Limitation Curve.

166 Scenario

167 Part of a Use Case. Splitting a Use Case into Scenarios helps readers understand the Use Case more
168 quickly. Some Scenarios are mandatory for a Use Case, whereas others may be recommended or
169 optional.

170 Specialization

171 Reusable data collection for a specific functionality.

172 SPINE

173 Smart Premises Interoperable Neutral-message Exchange: Technical Specification of EEBus Initiative
174 e.V.

175 TOUT

176 Short name of the Use Case "Time of Use Tariff"

177 UC

178 Abbreviation for "Use Case"

179 UTC

180 Abbreviation for "Coordinated Universal Time"

181

182 1.3 Requirements**183 1.3.1 Requirements wording**

184 The following keywords are used:

185 - SHALL

- 186 - SHALL NOT
- 187 - SHOULD
- 188 - SHOULD NOT
- 189 - MAY

190 Note: They apply only if written in capital letters.

191 For the meaning of the keywords, please refer to [RFC2119].

192

193 **1.3.2 Mapping of High-Level requirements**

194 Within the High-Level Use Case description, the following abbreviation is used:

195 [POEN-xyz]

196 e.g.: [POEN-007]

197 The abbreviation is used to mark High-Level requirements or rules of this Use Case with a unique
198 number xyz. These requirements are referenced throughout the technical solution to show how each
199 High-Level requirement is realized in the technical part.

200

2 High-Level description

2.1 Introduction

In smart electricity grids, it is important for the distribution system operator (DSO) to stabilize the grid at any time. As more and more regenerative energy production replaces traditional coal, gas and nuclear power plants, the production becomes less plannable. At the same time, the consumption is increased by high-power-consuming appliances like electric vehicles (EVs), heat pumps, etc.

One option for stabilizing the grid is to limit the consumption and production of the buildings connected to the DSO's electricity grid. For ad-hoc limitation, the Use Cases "Limitation of Power Consumption" (see section 2.5.1) and "Limitation of Power Production" (see section 2.5.2) are used. As the ad-hoc limitation must be followed, it may be problematic in regard to the appliance's goals (e.g., the EV's charging process, a heat pump's task to provide a desired room or warm water temperature, etc.). This Use Case ("Power Envelope") enables the DSO to send power limits in advance to prevent critical public grid situations, which in turn enables appliances to adjust their planning in advance.

The reference point for the power limitation is the building's grid connection point.

The Use Case "Power Envelope" provides power limits over time as a band of lower and upper limits for consumption and production (see Figure 1), sent by the Actor Transmission Broker (as the functional Actor for the DSO) to the Actor CEM ("Customer Energy Manager", as the functional Actor for the building's energy management system).

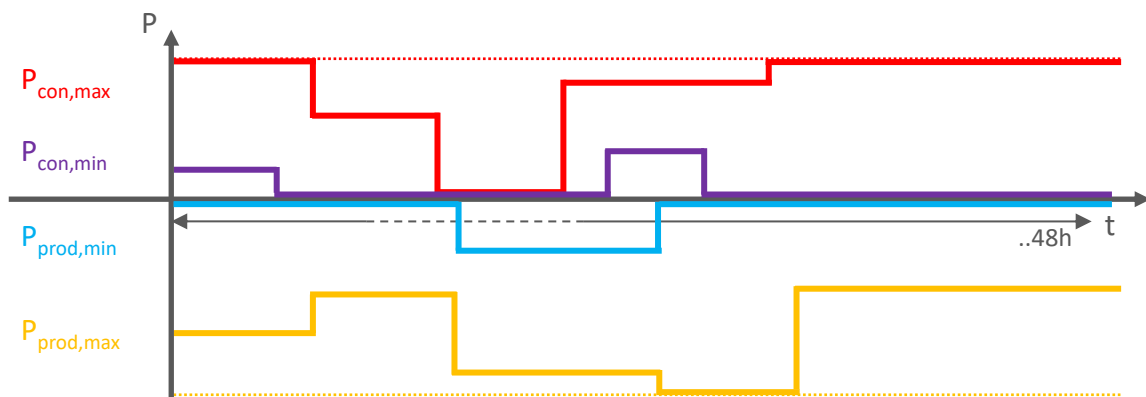


Figure 1: Power over time limit curves

The power limitation through this Use Case may trigger the Actor CEM to shift the load of connected appliances which may result in an updated power demand forecast (that is covered with the Use Case "Power Demand Forecast", see section 2.5.4).

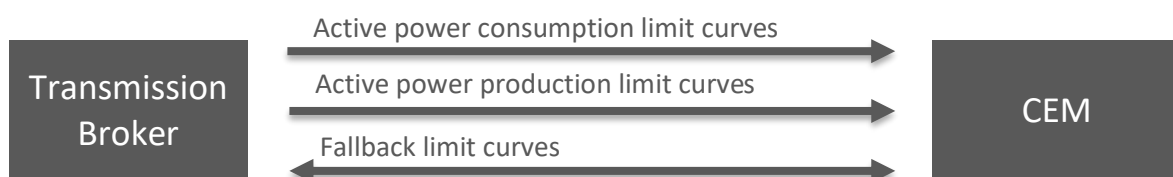


Figure 2: High-Level Use Case functionality overview

2.2 Detailed background information and rules

2.2.1 Overview and definitions

For consumption direction, this Use Case defines the following Limit Curves:

- Minimum Active Power Consumption Limit Curve ([POEN-011])
The power consumption of a building should not fall below this curve. See section 2.4.1 for details on the curve.
- Maximum Active Power Consumption Limit Curve ([POEN-012])
The power consumption of a building should not exceed this curve. See section 2.4.1 for details on the curve.
- Fallback Maximum Active Power Consumption Limit Curve ([POEN-031])
In particular cases this curve is used instead of the Maximum Active Power Consumption Limit Curve ([POEN-012]) to set an upper bound for the building's power consumption. See section 2.4.3 for details on the curve.

For production direction, this Use Case defines the following Limit Curves:

- Minimum Active Power Production Limit Curve ([POEN-021])
The power production of a building should not fall below the power production required by this curve (in terms of absolute power values). See section 2.4.2 for details on the curve.
- Maximum Active Power Production Limit Curve ([POEN-022])
The power production of a building should not exceed the power production required by this curve (in terms of absolute power values). See section 2.4.2 for details on the curve.
- Fallback Maximum Active Power Production Limit Curve ([POEN-032])
In particular cases this curve is used instead of the Maximum Active Power Production Limit Curve ([POEN-022]) to set a lower bound for the building's power production (in terms of absolute power values). See section 2.4.3 for details on the curve.

The curves partly share concepts and rules. Therefore, the following terms are used to associate the curves with the respective rules:

	Limit Curve	Power Limitation Curve	Power Envelope	Fallback Curve
Minimum Active Power Consumption Limit Curve ([POEN-011])	X	X	X	
Maximum Active Power Consumption Limit Curve ([POEN-012])	X	X		
Fallback Maximum Active Power Consumption Limit Curve ([POEN-031])	X			X
Minimum Active Power Production Limit Curve ([POEN-021])	X	X	X	
Maximum Active Power Production Limit Curve ([POEN-022])	X	X		
Fallback Maximum Active Power Production Limit Curve ([POEN-032])	X			X

Table 1: Terms used to associate curves with different rules.

Note: The term Power Envelope is used to express the combination of several Power Limitation Curves, set by the Transmission Broker.

Changes on the curves can be summarized as follows:

- The CEM enables the Transmission Broker to write one or more Power Limitations Curves. Which curves the CEM has to provide is defined within section 2.4.
- It is the responsibility of the Transmission Broker to decide if and which values it needs to write.
- During restart and runtime the CEM may remove or adjust some of these curves as explained in section 2.2.2.
- Power Limitation Curves have a finite duration, hence end at a given time. This means the Transmission Broker may need to update Power Limitation Curve(s) sometimes to maintain the validity^{*1} of the curves. More details can be found in section 2.2.2.2.

^{*1}: A Power Limitation Curve is "valid" if the curve has values for the current time and all values of the curve are within the permitted range (see section 2.6.2) and there are no gaps within the curve (see section 2.2.4).

2.2.2 System behaviour

Within this Use Case, the Actor CEM SHALL accept Limit Curves from at most one Actor Transmission Broker.

2.2.2.1 (Re)start behaviour

After any kind of restart, the CEM SHALL offer its Power Limitation Curves without any value (this implies that the CEM has either not stored previously written values persistently or has removed them during restart). Furthermore, the CEM SHALL NOT be limited by any Limit Curve of this Use Case. As soon as there is a Transmission Broker available that connected to the CEM and sent an initial Power Limitation Curve, the CEM SHALL begin to follow the Limit Curves as specified at the limitation behaviour (see section 2.2.2.2).

2.2.2.2 Limitation behaviour

As long as the CEM has valid Power Limitation Curves, the CEM SHALL apply the Power Limitation Curves to limit the building's power consumption or production according to the limitation obligations (see section 2.2.3.1). Please see below for an exception of this rule (switch to an unlimited behaviour).

The CEM MAY remove past slots.

If, later on, a Power Limitation Curve with associated Fallback Curve does not have values for the current time^{*1} (the last slot set by the Transmission Broker has ended), the according Fallback Curve

SHALL be applied (see section 2.2.6) according to the limitation obligations (see section 2.2.3.1) to limit the building's power consumption or production.

*¹ Note: This can happen if the curve has not been updated by the Transmission Broker in time, e.g., because the Transmission Broker is not available.

The application of a Fallback Curve can terminate in one of the following cases:

1. As soon as the Transmission Broker has written a valid Power Limitation Curve, the according Fallback Curve is not used by the CEM anymore to limit the building's consumption or production (see section 2.2.6 for details). Instead, the CEM SHALL apply the valid Power Limitation Curve again to limit the building's power consumption or production according to the limitation obligations (see section 2.2.3.1).
2. As soon as one of the Fallback Curves is applied for more than 24 hours (hence, there was no update on the according Power Limitation Curve for more than 24 hours), the CEM MAY manage the building's power consumption and production according to the unlimited behaviour*² (see section 2.2.2.3).

*²: This holds valid even if another Power Limitation Curve still has valid values. Example: The Transmission Broker sent Power Limitation Curves for consumption direction for 48 hours, but only for 12 hours for production direction. The Transmission Broker then doesn't send any further updates. After 12 hours, the Fallback Maximum Active Power Production Limit Curve ([POEN-032]) is applied for production direction (for consumption direction there are still valid values). 24 hours later, the unlimited behaviour is applied for the building's power consumption AND production (although there are still valid values for consumption direction). This correlates with the requirement from section 2.2.5 that at least once every 24 hours the Transmission Broker has to update the Power Limitation Curves. If the Transmission Broker does not act accordingly, this indicates that there might be an error at the Transmission Broker.

2.2.2.3 Unlimited behaviour

At the begin of the unlimited behaviour, the CEM SHALL delete all Power Limitation Curve values. During the unlimited behaviour, the building is not limited by any Limit Curve of this Use Case. Hence, the CEM MAY manage the building's appliances without limitations through this Use Case (other Use Cases may still limit the building's power consumption and production). As soon as the old or a new Transmission Broker sent a valid Power Limitation Curve, again, the building is limited by these Power Limitation Curve(s) again according to the limitation behaviour (see section 2.2.2.2).

2.2.3 General rules

2.2.3.1 Limitation obligations

The CEM SHALL follow*^{1,*2} the Power Limitation Curves set by the Transmission Broker. The same holds valid for the Fallback Curves, in case they are applied (please refer to section 2.2.6 for details).

*¹: Following the Limit Curves means that the CEM does its best effort to manage the controllable loads within the building in a way that the curves' limits are kept. The following reasons MAY prevent the CEM from achieving this:

- Uncontrolled loads or appliances that do not properly follow the CEM's commands to adjust their energy consumption or production.
 - Exceptions for the minimum power consumption or minimum power production as specified in the sections 2.4.1.1, 2.4.2.1 and 2.6.2.
 - Exceptions required through Use Cases with higher priorities, see section 2.5.1 and section 2.5.2.
- Note: This basically affects the minimum power consumption or minimum power production, not the maximum power consumption or maximum power production.

*²: Please refer to the end of section 2.2.2.2 for an exception of this rule (switch to an unlimited behaviour).

Note: The Transmission Broker should monitor the actual active power consumption or production of the CEM's building using the Use Case "Monitoring of Grid Connection Point" (see section 2.5.3).

2.2.3.2 Limit Curve persistence

As specified in section 2.2.2, the CEM's Power Limitation Curves are not used after a restart, hence there is no need to store them persistently.

Changes on the CEM's Fallback Curves MAY be stored persistently, although they will only be used if an existing connection to a Transmission Broker fails or the Transmission Broker misses to update the Power Limitation Curves (see section 2.2.6 for details). If the Fallback Curves are not stored persistently, the CEM will set its Fallback Curves to pre-configured default values.

2.2.4 Considered time horizon of Power Limitation Curves

The time horizon of the Power Limitation Curves SHOULD always cover at least the next 6 hours. Hence, latest 6 hours before the curve's time ends, an update SHOULD be sent to the CEM. This ensures that the CEM has enough time to manage the building's controllable appliances in a way that the limits can be kept.

If the CEM does not receive an update of the Power Limitation Curves before the last slot ends, the Fallback Curves become active (see section 2.2.6 for details).

The time period of a Power Limitation Curve SHALL start not later than the current time*¹ ("now") and end latest in 48 hours.

*¹: The following rules SHALL apply for the first slot of the Power Limitation Curve when it is received by the CEM:

- start time ≤ "now"
- start time + duration > "now"

Note: In case of inaccurate time synchronization, a receiving appliance could receive a start time of the first slot that lies in the future, or an end time (start time + duration) of the first slot that lies in the past (both are not allowed according to the rules above). However, the receiving device should calculate with a certain time tolerance*² in order not to reject an actually correct Power Limitation Curve as incorrect due to the inaccurate time synchronization.

*²: A tolerance of 5 minutes SHOULD be chosen.

As there should always be a Power Limitation Curve available that covers the next 6 hours, an update of the Power Limitation Curve(s) SHOULD cover at least the next 12 hours in order to not be required to send updates too often. Example: If an 8 hour Power Limitation Curve is sent, the next update should be sent latest after 2 hours. If a 12 hour Power Limitation Curve is sent, the next update should be sent latest after 6 hours.

The start time of each slot SHALL be expressed as UTC value with date information ([POEN-001/1]), whereas the slot's duration SHALL be presented as relative time ([POEN-001/3]). A Power Limitation Curve SHALL NOT have time gaps between two slots ([POEN-005/1]). A CEM SHALL reject a Power Limitation Curve that has time gaps.

Each slot SHALL have a duration of at least 5 minutes ([POEN-005/2]). A recommended minimum value for the duration is 15 minutes.

Note: All times are transmitted as UTC time. Hence, local times must be derived from that by the Actors themselves, if needed. It is strongly recommended that each Actor is "sufficiently synchronised" with the official "Coordinated Universal Time" (UTC).

Note: A Power Limitation Curve that covers a long time period may have a shorter time resolution (i.e. more slots) for the near future, while slots of the distant future have a coarser time resolution.

2.2.5 Updating the Power Envelope

As soon as the situation in the public electricity grid changes in a way that an update of the Power Envelope is needed, the Transmission Brokers sends a new Power Envelope to the CEM (with the time horizon requirements from section 2.2.4 fulfilled).

Latest every 24 hours the Transmission Broker SHALL update all valid Power Limitation Curve(s), even if the Transmission Broker does not change any value with the update.

An updated Power Limitation Curve always overwrites the existing data. Even if the old Power Limitation Curve has a longer time span than the new data, the old Power Limitation Curve will be completely overwritten with the new data. Hence, the CEM SHALL delete its existing Power Limitation Curve before storing the new data ([POEN-006]). This prevents the CEM from keeping outdated information within the Power Limitation Curve.

A Power Limitation Curve SHOULD NOT be updated more frequently than every 15 minutes. A CEM MAY reject a Power Limitation Curve that was sent within 15 minutes after the last Power Limitation Curve update.

2.2.6 Fallback Curves

In case one or more of the CEM's Power Limitation Curves run out of the last written time slot (i.e., they have no value for the current time), the CEM uses the according Fallback Curve(s) (see section 2.4.3) to limit the building's power consumption or production, respectively (as described in section 2.2.2.2). According to the rules described in section 2.2.2.2, this only holds valid for a certain time period.

The Fallback Maximum Active Power Consumption Limit Curve ([POEN-031]) is used by the CEM if the end time of the Maximum Active Power Consumption Limit Curve ([POEN-012]) is reached before the Transmission Broker updated the curve.

The Fallback Maximum Active Power Production Limit Curve ([POEN-032]) is used by the CEM if the end time of the Maximum Active Power Production Limit Curve ([POEN-022]) is reached before the Transmission Broker updated the curve.

There is no Fallback Curve for the Minimum Active Power Consumption Limit Curve ([POEN-011]) and the Minimum Active Power Production Limit Curve ([POEN-021]).

Both energy directions SHALL be considered separately. This means that, if (e.g.) the Maximum Active Power Consumption Limit Curve ([POEN-012]) still has a valid slot for the current time, but the Maximum Active Power Production Limit Curve ([POEN-022]) runs out of slots, only for production direction the Fallback Curve (in this example the Fallback Maximum Active Power Production Limit Curve ([POEN-032]) is used. Please refer to the end of section 2.2.2.2 for an exception of this rule (switch to an unlimited behaviour).

The CEM SHALL have pre-configured default Fallback Curves ([POEN-007]). One for the Fallback Maximum Active Power Consumption Limit Curve ([POEN-031]) and, if the data point is supported (see section 2.4 for details), one for the Fallback Maximum Active Power Production Limit Curve ([POEN-032]).

The Transmission Broker MAY change the CEM's Fallback Curves (the pre-configured default curves are then overwritten) ([POEN-008]).

The Fallback Curves SHALL be defined in the range from 00:00 to 24:00 (UTC)*³, without a specific date information ([POEN-001/2]). The Fallback Curves MAY have several slots or MAY have only one slot ([POEN-009]).

*³: For the majority of time zones a local time of 00:00:00 will be somewhere between the begin and end (00:00:00Z, 24:00:00Z) of a Fallback Curve.

The start time of each slot SHALL be expressed as UTC value ([POEN-001/1]) without date information ([POEN-001/2]), whereas the slot's duration SHALL be presented as relative time ([POEN-001/3]). A Fallback Curve SHALL NOT have time gaps between two slots ([POEN-005/1]). A CEM SHALL reject a Fallback Curve that has time gaps.

Each slot SHALL have a duration of at least 5 minutes ([POEN-005/2]). A recommended minimum value for the duration is 15 minutes.

Note: All times are transmitted as UTC time. Hence, local times must be derived from that by the Actors themselves, if needed.

2.2.7 Removing limitations

If the Transmission Brokers wants to remove a limitation of a given slot, the Transmission Broker can set the curve's slot to a value that effectively does not limit the building's power consumption or production^{*1}. In this case it is recommended to use the following values:

- For the Minimum Active Power Consumption Limit Curve ([POEN-011]): 0W
- For the Maximum Active Power Consumption Limit Curve ([POEN-012]): The according Slot Value Max ([POEN-013/7])
- For the Minimum Active Power Production Limit Curve ([POEN-021]): 0W
- For the Maximum Active Power Production Limit Curve ([POEN-022]): The according Slot Value Min ([POEN-023/6])^{*2}
- For the Fallback Maximum Active Power Consumption Limit Curve ([POEN-031]): The according Slot Value Max ([POEN-033/7])
- For the Fallback Maximum Active Power Production Limit Curve ([POEN-032]): The according Slot Value Min ([POEN-034/6])^{*2}

^{*1}: This principle also applies if the Transmission Broker updates a Limit Curve with new slots that shall not limit the building's power consumption or production. It may even be the case that a Limit Curve contains only slots with values that effectively do not impose a power limitation. In all cases, the recommendations help to avoid the occurrence of (forbidden) gaps in a curve.

^{*2}: For sign conventions see section 2.6.1.

Note: Please also consider section 2.6.2.

If the Transmission Broker requests the CEM to remove all slots of a Fallback Curve, the CEM SHALL accept the request and immediately afterwards set the Fallback Curve to pre-configured default values.

If there was one or more energy producing appliances in a building (which mandates the CEM to provide Scenario 2 with its Power Limitation Curves, but later all energy producing appliances are disconnected, the CEM SHALL delete the following data points as long as no energy producing appliance is available:

- The Power Limitation Curves [POEN-021] and [POEN-022]
- The Fallback Curve [POEN-032]
- The curves' constraints [POEN-023] and [POEN-034]

I.e., the CEM does not provide these Power Limitation Curves then (hence, the Transmission Broker is not able to write values for these curves).

2.3 Actors

2.3.1 Transmission Broker

The Actor Transmission Broker tries to stabilize the public electricity grid. Therefore it sends Power Limitation Curves to the buildings within the Transmission Broker's sub-part of the public grid, where needed.

2.3.2 CEM

The Actor CEM manages the devices in the building in order to prevent critical situations in the public electricity grid. The CEM receives Power Limitation Curves for the active power consumption and production (if supported).

2.4 Scenarios

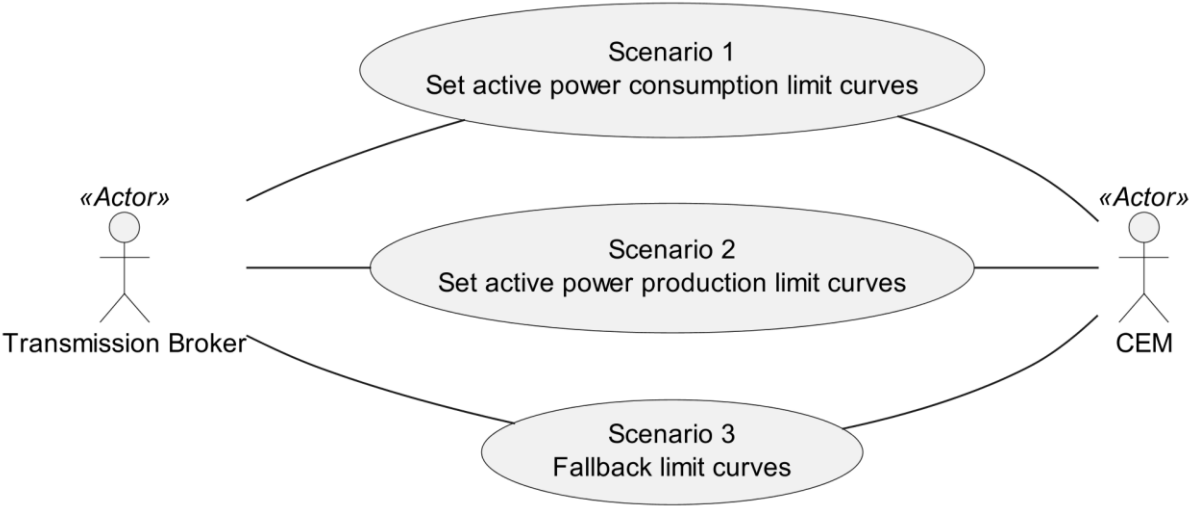


Figure 3: Scenario overview

Scenario number	Scenario name	Transmission Broker	CEM
1	Set active power consumption limit curves	M	M
2	Set active power production limit curves	M	F/M* ¹
3	Fallback limit curves	M	M

Table 2: Scenario implementation requirements for Actors

*¹: SHALL only be supported, if the CEM has connected appliances capable of producing energy. Otherwise the Scenario SHALL NOT be supported by the CEM.

2.4.1 Scenario 1 - Set active power consumption limit curves

2.4.1.1 Description

The Power Limitation Curves for the active (real) power consumption of the CEM are set with this Scenario.

The CEM SHALL confirm an accepted Power Limitation Curve (ACK, [POEN-003/1]). If the Power Limitation Curve cannot be applied by the CEM, the Transmission Broker SHALL be informed (NACK, [POEN-004/1]). In this case, the old Power Limitation Curve is not changed.

The following relation applies according to section 2.6.2:

- "Minimum Active Power Consumption Limit Curve"(t) ≤ "Maximum Active Power Consumption Limit Curve"(t)^{*2}:
At a given time t, the according value within the Minimum Active Power Consumption Limit Curve must be smaller or equal to the according value within the Maximum Active Power Consumption Limit Curve.

The Minimum Active Power Consumption Limit Curve ([POEN-011]) and the Maximum Active Power Consumption Limit Curve ([POEN-012]) SHALL share the same time slots.

The CEM SHALL state constraints of its supported active power consumption limit curves ([POEN-013]).

Data point name	Data point description [High-Level requirement]	Support indication	High-Level requirement
Minimum Active Power Consumption Limit Curve	Power over time curve stating the minimum active power the building SHALL consume.	M	[POEN-011] ^{*1,2}
Maximum Active Power Consumption Limit Curve	Power over time curve stating the maximum active power the building SHALL consume.	M	[POEN-012] ^{*2}
Active Power Consumption Limit Curve Constraints	Constraints for the CEM's active power consumption limit curves: - Slot Count Min ([POEN-013/1]) - Slot Count Max ([POEN-013/2])^{*4} - Slot Duration Min ([POEN-013/3]) - Slot Duration Max ([POEN-013/4]) - Slot Duration Step Size ([POEN-013/5]) - Slot Value Min ([POEN-013/6])^{*2,5} - Slot Value Max ([POEN-013/7])^{*2,6} - Slot Value Step Size ([POEN-013/8])^{*2}	M ^{*3}	[POEN-013]

Table 3: Scenario 1 - Set active power consumption limit curve - Data point list

^{*1}: A minimum active power consumption limit is not always applicable by the CEM, as there might not always be a controllable load that can be forced to consume power. Also, it should be avoided that the customer has a negative outcome of such a minimum limit as the building would consume more energy than actually needed for normal operation of the appliances.

^{*2}: For sign convention see section 2.6.1.

^{*3}: The data points Slot Count Max ([POEN-013/2]), Slot Value Min ([POEN-013/6]) and Slot Value Max ([POEN-013/7]) SHALL be supported by the CEM. Further constraints MAY be supported by the CEM.

^{*4}: The Actor CEM SHALL support at least a Slot Count Max ([POEN-013/2]) of 192 (higher values are always possible).

^{*5}: The value for Slot Value Min ([POEN-013/6]) SHALL be 0W.

*⁶: It is recommended to set the value for Slot Value Max ([POEN-013/7]) to the value of the data point Contractual Consumption Nominal Max ([PODF-021], see section 2.5.4). Please refer to section 2.6.2 for details.

2.4.1.2 Conditions

Triggering Event:

The Actor Transmission Broker wants to update the active power consumption limit curves of the Actor CEM.

Pre-condition:

The active power consumption limit curves of the Actor CEM are outdated.

Post-condition:

The Actor CEM has updated active power consumption limit curves.

2.4.2 Scenario 2 - Set active power production limit curves

2.4.2.1 Description

The Power Limitation Curves for the active (real) power production of the CEM are set with this Scenario.

The CEM SHALL confirm an accepted Power Limitation Curve (ACK, [POEN-003/2]). If the Power Limitation Curve cannot be applied by the CEM, the Transmission Broker SHALL be informed (NACK, [POEN-004/2]). In this case, the old Power Limitation Curve is not changed.

The following relation applies according to section 2.6.2:

- "Minimum Active Power Production Limit Curve"(t) $\geq$ "Maximum Active Power Production Limit Curve"(t)*²:
At a given time t, the according value within the Minimum Active Power Production Limit Curve must be (mathematically) greater or equal to the according value within the Maximum Active Power Production Limit Curve (hence, the absolute value of the Minimum Active Power Production Limit Curve must be smaller or equal to the absolute value of the Maximum Active Power Production Limit Curve).

The terms "Minimum" and "Maximum" of the curve names in this section ([POEN-021] and [POEN-022]) refer to the generator view with positive energy feed-in, whereas the mathematical relation of the curves' values applied in this Use Case follow the passive sign convention*². As described above, the value at a given time t of the "Minimum" curve ([POEN-021]) shall be mathematically greater than the "Maximum" curve ([POEN-022]). "Normally", when interpreting values in a mathematical way, a minimum is smaller than a maximum. This must be kept in mind when implementing this Use Case.

The Minimum Active Power Production Limit Curve ([POEN-021]) and the Maximum Active Power Production Limit Curve ([POEN-022]) SHALL share the same time slots.

560 The CEM SHALL state constraints of its supported active power production limit curves ([POEN-023]).

Data point name	Data point description [High-Level requirement]	Support indication	High-Level requirement
Minimum Active Power Production Limit Curve	Power over time curve stating the minimum active power the building SHALL produce.	M	[POEN-021]* ^{1,2}
Maximum Active Power Production Limit Curve	Power over time curve stating the maximum active power the building SHALL produce.	M	[POEN-022]* ²
Active Power Production Limit Curve Constraints	Constraints for the CEM's active power production limit curves: - Slot Count Min ([POEN-023/1]) - Slot Count Max ([POEN-023/2])*⁴ - Slot Duration Min ([POEN-023/3]) - Slot Duration Max ([POEN-023/4]) - Slot Duration Step Size ([POEN-023/5]) - Slot Value Min ([POEN-023/6])*^{2,5} - Slot Value Max ([POEN-023/7])*^{2,6} - Slot Value Step Size ([POEN-023/8])*²	M* ³	[POEN-023]

561 Table 4: Scenario 2 - Set active power production limit curve - Data point list

562 *¹: A minimum active power production limit is not always applicable by the CEM, as there might not
563 always be a controllable appliance that can be forced to produce power. Also, it should be avoided
564 that the customer has a negative outcome of such a minimum production limit as the building would
565 feed more power into the grid instead of consuming it, which may later on result in a higher grid-
566 consumption at typically higher costs.

567 *²: For sign convention see section 2.6.1.

568 *³: The data point Slot Count Max ([POEN-023/2]), Slot Value Min ([POEN-023/6]) and Slot Value Max
569 ([POEN-023/7]) SHALL be supported by the CEM. Further constraints MAY be supported by the CEM.

570 *⁴: The Actor CEM SHALL support at least a Slot Count Max ([POEN-023/2]) of 192 (higher values are
571 always possible).

572 *⁵: It is recommended to set the value for Slot Value Min ([POEN-023/6]) to "-(Contractual
573 Production Nominal Max)" ([PODF-022], see section 2.5.4). Please refer to section 2.6.2 for details.

574 *⁶: The value for Slot Value Max ([POEN-023/7]) SHALL be 0W.

575

576 2.4.2.2 Conditions

577 Triggering Event:

578 The Actor Transmission Broker wants to update the active power production limit curves of the Actor
579 CEM.

580 Pre-condition:

581 The active power production limit curves of the Actor CEM are outdated.

Post-condition:

The Actor CEM has updated active power production limit curves.

2.4.3 Scenario 3 - Fallback limit curves**2.4.3.1 Description**

The Fallback Maximum Active Power Consumption Limit Curve ([POEN-031]) and the Fallback Maximum Active Power Production Limit Curve ([POEN-032]) are used by the CEM as limiting curves if there is no according valid Power Limitation Curve given for the current time.

This may occur (e.g.) if the communication between the Transmission Broker and the CEM is broken. To prevent an unknown behaviour of the CEM in such cases, the Fallback Curves are applied by the CEM.

The CEM SHALL confirm an accepted Fallback Curve (ACK, [POEN-003/3]). If the Fallback Curve cannot be applied by the CEM, the Transmission Broker SHALL be informed (NACK, [POEN-004/3]). In this case, the old Fallback Curve is not changed.

For more details and rules on the Fallback Curves, please refer to section 2.2.6.

The CEM SHALL state constraints of its supported Fallback Curves ([POEN-033] and [POEN-034]).

Data point name	Data point description [High-Level requirement]	Support indication	High-Level requirement
Fallback Maximum Active Power Consumption Limit Curve	This Fallback Curve applies if the CEM has no valid Maximum Active Power Consumption Limit Curve ([POEN-012]) for the current time.	M	[POEN-031]* ²
Fallback Maximum Active Power Production Limit Curve	This Fallback Curve applies if the CEM has no valid Maximum Active Power Production Limit Curve ([POEN-022]) for the current time.	F/M* ¹	[POEN-032]* ²
Fallback Active Power Consumption Limit Curve Constraints	Constraints for the CEM's Fallback Maximum Active Power Consumption Limit Curve: - Slot Count Min ([POEN-033/1]) - Slot Count Max ([POEN-033/2])*⁵ - Slot Duration Min ([POEN-033/3]) - Slot Duration Max ([POEN-033/4]) - Slot Duration Step Size ([POEN-033/5]) - Slot Value Min ([POEN-033/6])*^{2,7} - Slot Value Max ([POEN-033/7])*^{2,8} - Slot Value Step Size ([POEN-033/8])*²	M* ³	[POEN-033]
Fallback Active Power Production Limit Curve Constraints	Constraints for the CEM's Fallback Maximum Active Power Production Limit Curve: - Slot Count Min ([POEN-034/1]) - Slot Count Max ([POEN-034/2])*⁶ - Slot Duration Min ([POEN-034/3]) - Slot Duration Max ([POEN-034/4]) - Slot Duration Step Size ([POEN-034/5]) - Slot Value Min ([POEN-034/6])*^{2,9}	F/M* ^{1,4}	[POEN-034]

	- Slot Value Max ([POEN-034/7])* ^{2,10}		
	- Slot Value Step Size ([POEN-034/8])* ²		

Table 5: Scenario 3 - Fallback limit curves - Data point list

*¹: This data point SHALL only be supported by the Actor CEM, if the CEM supports the Scenario "Set active power production limit curve".

*²: For sign convention see section 2.6.1.

*³: The data points Slot Count Max ([POEN-033/2]), Slot Value Min ([POEN-033/6]) and Slot Value Max ([POEN-033/7]) SHALL be supported by the CEM. Further constraints MAY be supported by the CEM.

*⁴: The data points Slot Count Max ([POEN-034/2]), Slot Value Min ([POEN-034/6]) and Slot Value Max ([POEN-034/7]) SHALL be supported by the CEM. Further constraints MAY be supported by the CEM.

*⁵: The Actor CEM SHALL support at least a Slot Count Max ([POEN-033/2]) of 96 (higher values are always possible).

*⁶: The Actor CEM SHALL support at least a Slot Count Max ([POEN-034/2]) of 96 (higher values are always possible).

*⁷: The value for Slot Value Min ([POEN-033/6]) SHALL be 0W.

*⁸: It is recommended to set the value for Slot Value Max ([POEN-033/7]) to the value of the data point Contractual Consumption Nominal Max ([PODF-021], see section 2.5.4). Please refer to section 2.6.2 for details.

*⁹: It is recommended to set the value for Slot Value Min ([POEN-034/6]) to "-(Contractual Production Nominal Max)" ([PODF-022], see section 2.5.4). Please refer to section 2.6.2 for details.

*¹⁰: The value for Slot Value Max ([POEN-034/7]) SHALL be 0W.

2.4.3.2 Conditions

Triggering Event:

The Actor Transmission Broker is interested in the Fallback Curves of the Actor CEM or wants to change the Fallback Curves.

Pre-condition:

The Actor Transmission Broker does not know the Fallback Curves of the Actor CEM or the Fallback Curves are outdated.

Post-condition:

The Actor Transmission Broker knows the Fallback Curves of the Actor CEM and the Fallback Curves are up-to-date.

2.5 Dependencies to other Use Cases

2.5.1 "Limitation of Power Consumption"

The Use Case "Limitation of Power Consumption" (LPC) is used to limit the power consumption of an appliance or a building^{*1} (by the use of a CEM).

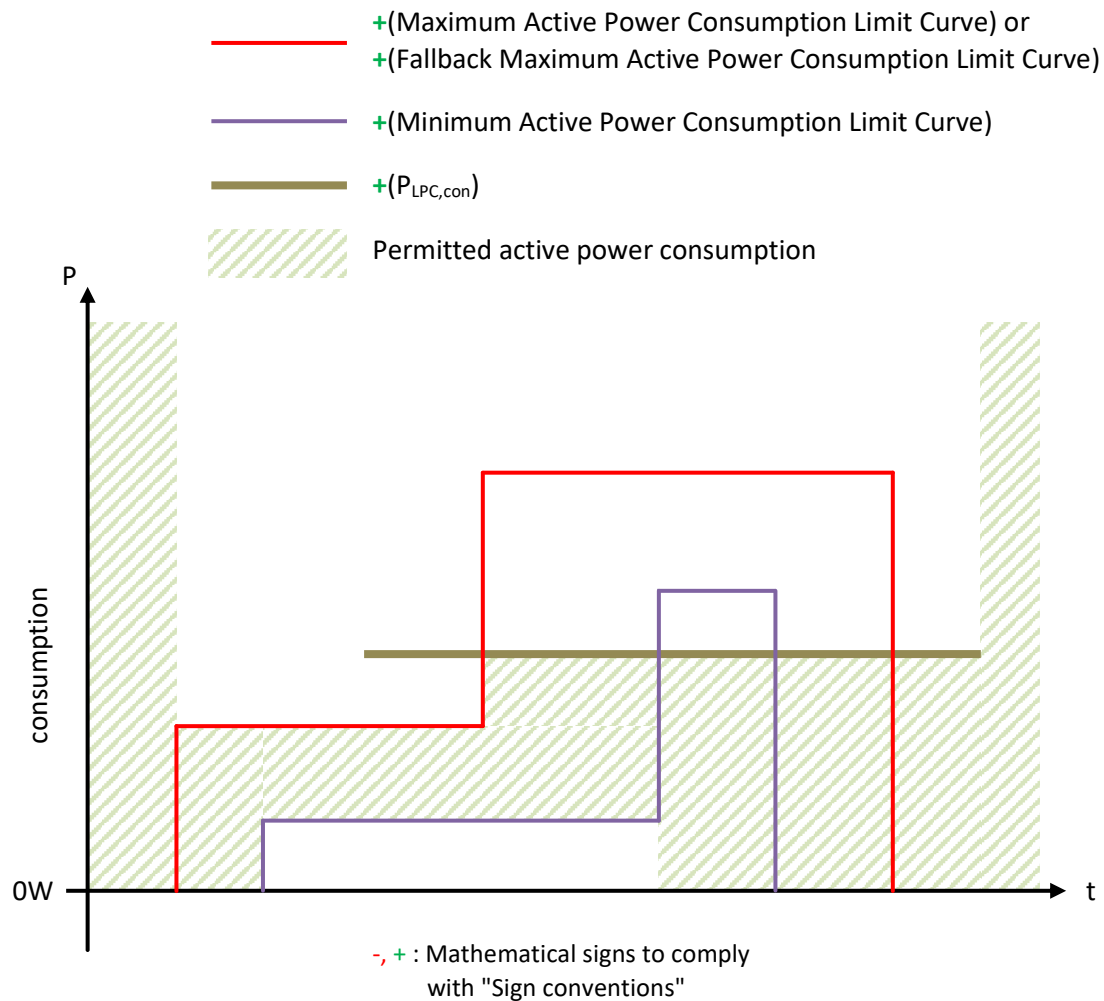
^{*1}: Please consider country specific regulations regarding the application scope of the Use Case LPC.

If the Actor CEM supports the Use Case "Limitation of Power Consumption" and this Use Case, the Actor CEM SHALL only be controlled by the same logical instance. This means, that both, the Actor Energy Guard of the Use Case LPC and the Actor Transmission Broker of this Use Case SHALL be implemented on the same logical instance.

If the Actor CEM supports the Use Case "Limitation of Power Consumption", the following applies^{*2}:

- The Use Case "Limitation of Power Consumption" describes power consumption limits like the Active Power Consumption Limit ([LPC-011]) or the Failsafe Consumption Active Power Limit ([LPC-021]). It also specifies which of the values is applied and if the limits are activated or not. In Figure 4, "P_{LPC,con}" represents a power consumption limit from the Use Case LPC. It is only shown for situations when this limit is activated according to the LPC UC.
- The Actor CEM limits the building's power consumption using rules and values of the Maximum Active Power Consumption Limit Curve ([POEN-012]) and the Fallback Maximum Active Power Consumption Limit Curve ([POEN-031]). However, the Actor CEM SHALL NOT violate a power consumption limitation imposed by the LPC UC.
Note: This means that the Actor CEM's power consumption might be limited by the LPC UC and may be even further limited by [POEN-012] or [POEN-031].
- The Actor CEM may try to achieve a minimum building power consumption using rules and values of the Minimum Active Power Consumption Limit Curve ([POEN-011]). However, the Actor CEM SHALL NOT violate a power consumption limitation imposed by the LPC UC.
Note: This means that the Actor CEM's power consumption might be limited by the LPC UC in a way that the Actor CEM is not obliged to consume power at all if a consumption limit from the LPC UC is too low to achieve a minimum consumption.

^{*2}: For sign conventions see section 2.6.1.



659

660 *Figure 4: Example of permitted active power consumption (hatched area)*

661 The Actor Transmission Broker SHOULD support the Use Case "Limitation of Power Consumption".

662 The Actor CEM SHOULD support the Use Case "Limitation of Power Consumption".

663 The Actor Transmission Broker of this Use Case acts as Actor Energy Guard within the Use Case
664 "Limitation of Power Consumption".665 The Actor CEM of this Use Case acts as Actor Controllable System within the Use Case "Limitation of
666 Power Consumption".

667

668 **2.5.2 "Limitation of Power Production"**669 The Use Case "Limitation of Power Production" (LPP) is used to limit the power production of an
670 appliance or a building*¹ (by the use of a CEM).671 *¹: Please consider country specific regulations regarding the application scope of the Use Case LPP.672 If the Actor CEM supports the Use Case "Limitation of Power Production", the following applies*²:

- The Use Case "Limitation of Power Production" describes power production limits like the Active Power Production Limit ([LPP-011]) or the Failsafe Production Active Power Limit ([LPP-021]). It also specifies which of the values is applied and if the limits are activated or not.
In Figure 5, " $P_{LPP,prod}$ " represents a power production limit from the Use Case LPP. It is only shown for situations when this limit is activated according to the LPP UC.
- The Actor CEM limits the building's power production using rules and values of the Maximum Active Power Production Limit Curve ([POEN-022]) and the Fallback Maximum Active Power Production Limit Curve ([POEN-032]). However, the Actor CEM SHALL NOT violate a power production limitation imposed by the LPP UC.
Note: This means that the Actor CEM's power production might be limited by the LPP UC and may be even further limited by [POEN-022] or [POEN-032].
- The Actor CEM may try to achieve a minimum building power production using rules and values of the Minimum Active Power Production Limit Curve ([POEN-021]). However, the Actor CEM SHALL NOT violate a power consumption limitation imposed by the LPP UC.
Note: This means that the Actor CEM's power production might be limited by the LPP UC in a way that the Actor CEM is not obliged to produce power at all if a production limit from the LPP UC is too low to achieve a minimum production.

*2: For sign conventions see section 2.6.1.

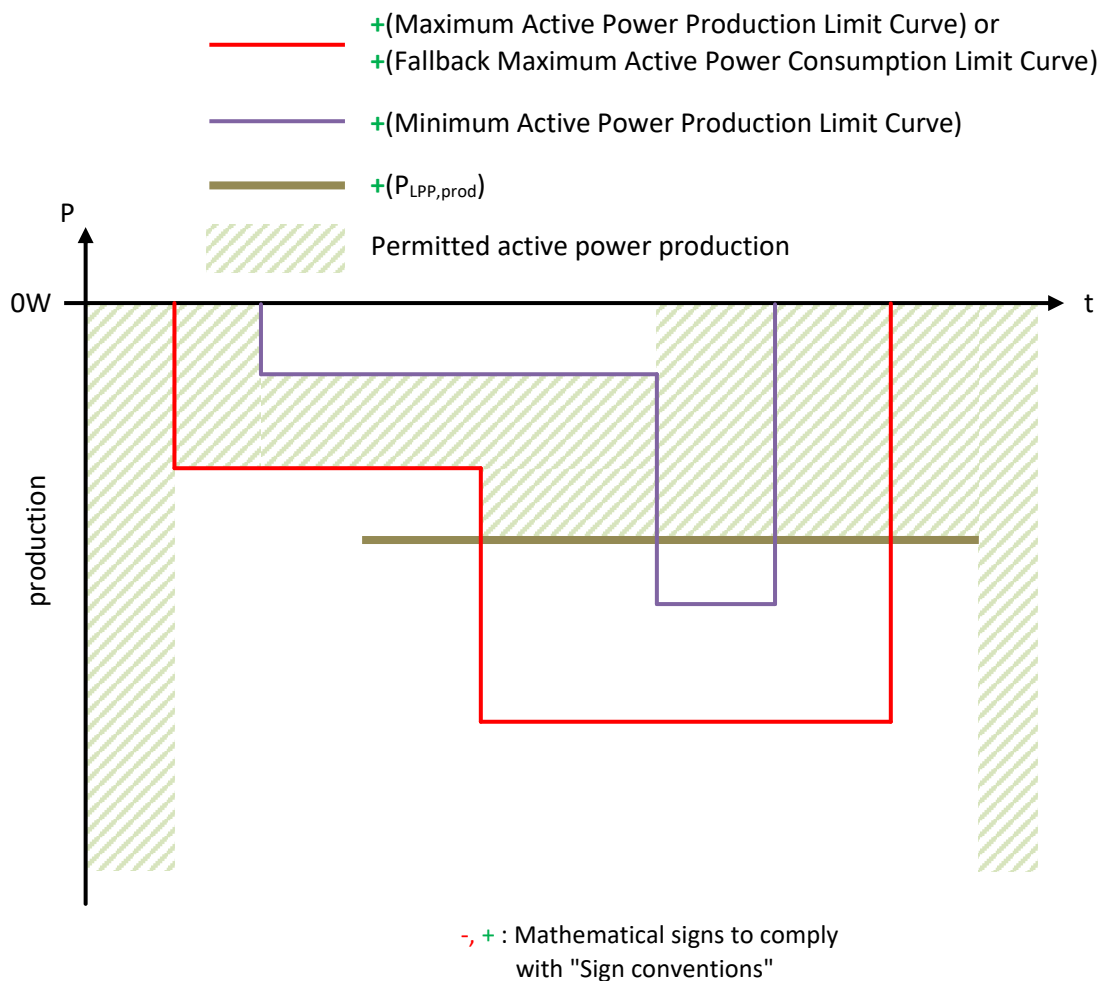


Figure 5: Example of permitted active power production (hatched area)

693 The Actor Transmission Broker SHOULD support the Use Case "Limitation of Power Production".

694 The Actor CEM SHOULD support the Use Case "Limitation of Power Production".

695 The Actor Transmission Broker of this Use Case acts as Actor Energy Guard within the Use Case
696 "Limitation of Power Production".

697 The Actor CEM of this Use Case acts as Actor Controllable System within the Use Case "Limitation of
698 Power Production".

699

700 **2.5.3 "Monitoring of Grid Connection Point"**

701 This dependency only holds valid for the Actor Transmission Broker and is only for informative
702 purposes.

703 The Use Case "Monitoring of Grid Connection Point" enables the Transmission Broker to monitor the
704 power consumption or production at the Grid Connection Point.

705 The Actor Transmission Broker SHOULD support the Use Case "Monitoring of Grid Connection Point".

706 The Actor Transmission Broker of this Use Case acts as Actor Monitoring Appliance within the Use
707 Case "Monitoring of Grid Connection Point".

708

709 **2.5.4 "Power Demand Forecast"**

710 The Use Case "Power Demand Forecast" is used by the Actor CEM to provide information about its
711 planned power consumption / production. Within a power demand forecast, the CEM can send
712 multiple time slots, each with the estimated (planned) power consumption / production. The
713 information can be sent to all interested Actors, e.g. the Actor Transmission Broker.

714 The Use Case "Power Demand Forecast" provides (among others) the two data points

715 - Contractual Consumption Nominal Max ([PODF-021])

716 - Contractual Production Nominal Max ([PODF-022])

717 that are used for the relations given in section 2.6.1.

718 Note: Both data points can be retrieved via the Use Cases "Limitation of Power Consumption" and
719 "Limitation of Power Production" as well (Contractual Consumption Nominal Max ([LPC-042]) via LPC
720 and Contractual Production Nominal Max ([LPP-042]) via LPP).

721 An updated power demand forecast of the Actor CEM MAY trigger the Actor Transmission Broker to
722 send new limits to the Actor CEM with this Use Case.

723 The Actor Transmission Broker SHOULD support the Use Case "Power Demand Forecast".

724 The Actor CEM SHOULD support the Use Case "Power Demand Forecast".

725 The Actor Transmission Broker of this Use Case acts as Actor Transmission Broker within the Use
726 Case "Power Demand Forecast".

727 The Actor CEM of this Use Case acts as Actor CEM within the Use Case "Power Demand Forecast".

728

729 2.5.5 "Time of Use Tariff"

730 The Use Case "Time of Use Tariff" is used by the Actor Transmission Broker to inform the Actor CEM
731 about the upcoming incentive (e.g. price) values for the CEM's power consumption or production
732 (transmission fee only).

733 The Actor Transmission Broker SHOULD support the Use Case "Time of Use Tariff".

734 The Actor CEM SHOULD support the Use Case "Time of Use Tariff".

735 The Actor Transmission Broker of this Use Case acts as Actor Transmission Broker within the Use
736 Case "Time of Use Tariff".

737 The Actor CEM of this Use Case acts as Actor CEM within the Use Case "Time of Use Tariff".

738

739 2.6 Further information and rules

740 2.6.1 Sign conventions

741 The following values of this Use Case follow the passive sign convention (positive values indicate
742 consumption, negative values indicate production) ([POEN-002/1]):

- 743 - Minimum Active Power Consumption Limit Curve ([POEN-011])
- 744 - Maximum Active Power Consumption Limit Curve ([POEN-012])
- 745 - Active Power Consumption Limit Curve Constraints: Slot Value Min ([POEN-013/6])
- 746 - Active Power Consumption Limit Curve Constraints: Slot Value Max ([POEN-013/7])
- 747 - Minimum Active Power Production Limit Curve ([POEN-021])
- 748 - Maximum Active Power Production Limit Curve ([POEN-022])
- 749 - Active Power Production Limit Curve Constraints: Slot Value Min ([POEN-023/6])
- 750 - Active Power Production Limit Curve Constraints: Slot Value Max ([POEN-023/7])
- 751 - Fallback Maximum Active Power Consumption Limit Curve ([POEN-031])
- 752 - Fallback Maximum Active Power Production Limit Curve ([POEN-032])
- 753 - Fallback Active Power Consumption Limit Curve Constraints: Slot Value Min ([POEN-033/6])
- 754 - Fallback Active Power Consumption Limit Curve Constraints: Slot Value Max ([POEN-033/7])
- 755 - Fallback Active Power Production Limit Curve Constraints: Slot Value Min ([POEN-034/6])
- 756 - Fallback Active Power Production Limit Curve Constraints: Slot Value Max ([POEN-034/7])

757 The following values of this Use Case are always positive ([POEN-002/2]):

- 758 - Active Power Consumption Limit Curve Constraints: Slot Value Step Size ([POEN-013/8])
- 759 - Active Power Production Limit Curve Constraints: Slot Value Step Size ([POEN-023/8])
- 760 - Fallback Active Power Consumption Limit Curve Constraints: Slot Value Step Size ([POEN-
761 033/8])
- 762 - Fallback Active Power Production Limit Curve Constraints: Slot Value Step Size ([POEN-
763 034/8])

2.6.2 Permitted ranges for Limit Curves

The following values that are not part of this Use Case are relevant for the subsequent relations (see section 2.5.4):

- Contractual Consumption Nominal Max: The nominal maximum active (real) power the building is allowed to consume due to the customer's contract.
- Contractual Production Nominal Max: The nominal maximum active (real) power the building is allowed to produce due to the customer's contract.

Both values are always greater than or equal to zero. Their name indicates whether they are used for consumption or production direction (see section "Sign conventions" within the Use Case "Power Demand Forecast"):

- Contractual Consumption Nominal Max: consumption direction
- Contractual Production Nominal Max: production direction

If the CEM also supports the Use Case "Power Demand Forecast" (see section 2.5.4) and the data points Contractual Consumption Nominal Max ([PODF-021]) or Contractual Production Nominal Max ([PODF-022]), the CEM

- SHALL NOT consume more power than permitted by the Contractual Consumption Nominal Max ([PODF-021]),
- SHALL NOT produce more (absolute) power than permitted by the Contractual Production Nominal Max ([PODF-022]),
- SHALL NOT reject a Limit Curve that exceeds the limit provided by Contractual Consumption Nominal Max ([PODF-021]) or Contractual Production Nominal Max ([PODF-022]), resp.*¹

*¹: The Transmission Broker might not support the data points [PODF-021] and [PODF-022] and therefore could send Limit contradicting the contractual limitations. If the Transmission Broker supports the data points [PODF-021] and [PODF-022], it SHALL NOT send Limit Curves contradicting the contractual limitations.

Furthermore, the following relations SHALL be kept for every time t:

- $0 = \text{Slot Value Min ([POEN-013/6])} \leq \text{"Minimum Active Power Consumption Limit Curve"(t)} \leq \text{"Maximum Active Power Consumption Limit Curve"(t)} \leq \text{Slot Value Max ([POEN-013/7])}$
- $0 = \text{Slot Value Max ([POEN-023/7])} \geq \text{"Minimum Active Power Production Limit Curve"(t)} \geq \text{"Maximum Active Power Production Limit Curve"(t)} \geq \text{Slot Value Min ([POEN-023/6])}$
- $0 = \text{Slot Value Min ([POEN-033/6])} \leq \text{"Fallback Maximum Active Power Consumption Limit Curve"(t)} \leq \text{Slot Value Max ([POEN-033/7])}$
- $0 = \text{Slot Value Max ([POEN-034/7])} \geq \text{"Fallback Maximum Active Power Production Limit Curve"(t)} \geq \text{Slot Value Min ([POEN-034/6])}$

If the CEM receives a Limit Curve that contradicts the rule described above, the CEM SHALL reject the Limit Curve.

The following figure shows a simplified example of the permitted ranges for the Power Limitation Curves:

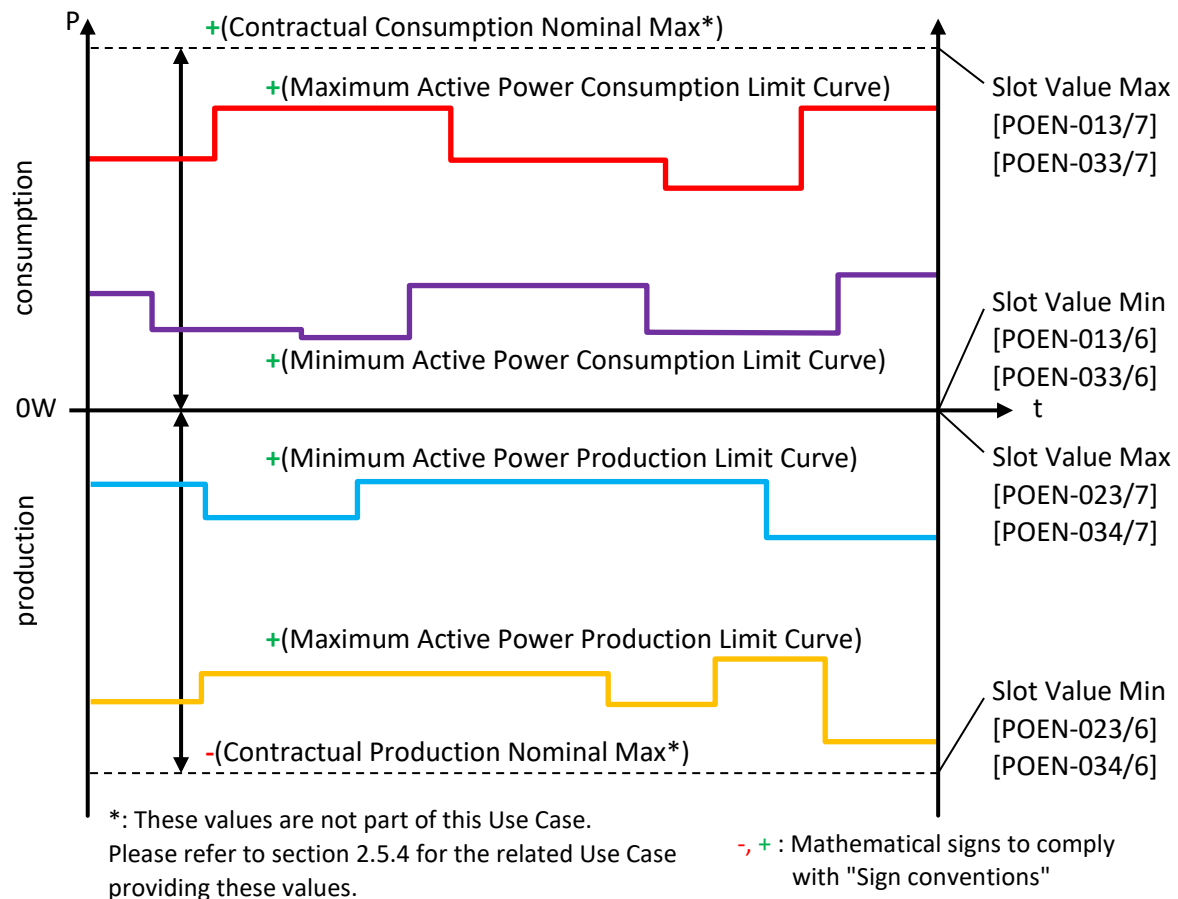


Figure 6: Example of permitted ranges for power limit curves

Note: The Slot Value Max values [POEN-013/7] and [POEN-033/7] can differ from the Contractual Consumption Nominal Max value as well as from each other.

Note: The Slot Value Min values [POEN-023/6] and [POEN-034/6] can differ from the Contractual Production Nominal Max value as well as from each other.

The permitted range of operation for the building is then:

- For consumption direction:
 - Between Minimum Active Power Consumption Limit Curve and Maximum Active Power Consumption Limit Curve
- For production direction (please consider the sign conventions in section 2.6.1):
 - Between Minimum Active Power Production Limit Curve and Maximum Active Power Production Limit Curve

As soon as the Minimum Active Power Consumption Limit Curve or the Minimum Active Power Production Limit Curve (or both) are set to values different than "0W", there is a gap within the power range that the building is not permitted to be operated in.

Use Cases with higher priority may impose stronger power limitations and supersede the above-mentioned ranges (e.g., the Use Cases LPC and LPP, see sections 2.5.1 and 2.5.2).

822 Note: As described in section 2.4.1 and 2.4.2, it is not always possible to follow the Minimum Active
823 Power Consumption Limit Curve and the Minimum Active Power Production Limit Curve.

824

825 **2.7 Assumptions and Prerequisites**

826 None.

827

3 Technical SPINE solution

3.1 General rules and information

3.1.1 Underlying technology documents

This technical solution relies on the SPINE Resources Specification version 1.4.0 [ResourceSpecification].

For interoperable connectivity this technical solution relies on:

- SPINE Protocol Specification version 1.4.0 [ProtocolSpecification] as application protocol.
- SHIP Specification version 1.0.1 [SHIP] as transport protocol.

3.1.2 Use Case discovery rules

Use Case discovery SHALL be supported by each Actor and the following rules SHALL apply:

- The string content for the Element "nodeManagementUseCaseData. useCaseInformation. useCaseSupport. useCaseName" within the Use Case discovery (please refer to [ProtocolSpecification]) SHALL be "powerEnvelope". The string content SHALL only be defined by this Use Case (regardless of the Use Case version).
- The string content of the Element "nodeManagementUseCaseData. useCaseInformation. actor" within the Use Case discovery (please refer to [ProtocolSpecification]) SHALL be set to the according value stated within the corresponding Actor's section.
- An Actor A that is implemented to support this Use Case specification SHALL set the Element "nodeManagementUseCaseData. useCaseInformation. useCaseSupport. useCaseVersion" within the Use Case discovery (please refer to [ProtocolSpecification]) to "1.0.0".
- If an Actor A supports multiple versions of this Use Case with the same major version number, only the highest one SHOULD be set within the Use Case discovery.
- If an Actor A finds a proper counterpart Actor B for this Use Case that supports multiple versions of this Use Case with the same major version number as supported by Actor A, the Actor A SHOULD evaluate from these versions of Actor B only the highest version number.
- If an Actor A supports multiple versions of this Use Case with different major version numbers, for each major version number only the highest version number SHOULD be set within the Use Case discovery.
- If an Actor A finds a proper counterpart Actor B for this Use Case that supports only versions with a major version number not implemented by Actor A, it still might be possible to run the Use Case or parts of the Use Case. Therefore, the Actor A should try to evaluate the Actor B as a valid partner for this Use Case.

3.1.3 Rules for "Content of Specialization..." tables and "Content of Function..." tables

3.1.3.1 General presence indication definitions

Abbreviations for the presence indication of Elements listed in the tables are defined as follows:

Abbreviation	Meaning	Link to requirement keywords
--------------	---------	------------------------------

M	Mandatory	SHALL
R	Recommended	SHOULD
O	Optional	MAY

Table 6: Presence indication description

An Actor MAY support Elements that are not listed in the tables. However, another Actor MAY ignore these Elements.

The presence indications "M", "R" and "O" are always meant relative to the respective parent Element. I.e. if a parent Element is optional ("O") and a child is mandatory ("M") the child Element can only be present if the parent Element is present as well.

Note: The indications and the aforementioned rules apply for "complete messages" (so-called "full function exchange", please refer to [ProtocolSpecification]). In contrast, the so-called "restricted function exchange" is designed to permit exchange of specific excerpts of data, i.e. fewer Elements than potentially available from the data owner (partially even not all "mandatory" Elements).

3.1.3.2 Presence indications for "Content of Specialization..." tables

This section only defines rules for the client side.

Elements that are marked with "M" SHALL be supported by the client in case of readable as well as writeable data. This Element may be optional on the server side.

The following applies for readable data that is exchanged in a "read/reply" or "notify" operation:

- "R" means that the data SHOULD be supported by the client. In other words: If the server responds with the according Element, the client SHOULD be able to interpret the according Elements.
- "O" means that the data MAY be supported by the client. In other words: If the server responds with the according Element, the client MAY be able to interpret the according Elements.

The following applies for writeable data that is exchanged in a "write" operation:

- "R" means that the data SHOULD be written by the client.
- "O" means that the data MAY be written by the client.
- "F" means that the data SHALL NOT be written by the client.

The following applies for Elements that are not listed in the Actor section:

- In case of a received "reply" message: The client MAY ignore the Element.
- In case of a "write" operation to be created: The client MAY set the Element but SHALL consider that the server may ignore the Element.
- In case of a received "notify" message: The client MAY ignore the Element.

M, R or O may be combined with the suffix "(event)" to express that a supported Element or value only has to be supported during a certain event and hence does not need to be present at all times. If the event is not active the Element may be omitted or another value may be set. In most cases a High-Level requirement reference for the event is given in the rules column.

900 In some cases, an Element may have a double presence indication, separated by a comma. Examples:

- 901 - M, O
- 902 - M \W, O \W

903 The description (rules) of the Element details in such a case the presence requirement. A typical
904 example is a list-based Element where the "first" (or "last") Element has a different presence
905 requirement than the other list Elements.

906

907 **3.1.3.3 Presence indications for "Content of Function..." tables**

908 This section only defines rules for the server side.

909 Elements that are marked with "M" SHALL be supported by the server in case of readable as well as
910 writeable data. In case of writeable data (marked with "M \W") the server does not need to set the
911 Element, because the Element is set only by the client.

912 The following applies for readable data that is exchanged in a "read/reply" or "notify" operation:

- 913 - "R" means that the data SHOULD be provided by the server.
- 914 - "O" means that the data MAY be provided by the server.
- 915 - "F" means that the data SHALL NOT be provided by the server.

916 The following applies for writeable data that is exchanged in a "write" operation:

- 917 - "R" means that the data SHOULD be supported. In other words: If the client writes the
918 Element, the server SHOULD accept those messages and the contained Elements.
- 919 - "O" means that the data MAY be supported. In other words: If the client writes the Element,
920 the server MAY accept those messages and the contained Elements.

921 The following applies for Elements that are not listed in the Actor section:

- 922 - In case of a received "read" request: The according Element MAY be set in the reply.
- 923 - In case of a received "write" operation: The server MAY ignore the Element.
- 924 - In case of a "notify" operation to be created: The server MAY set the Element.

925 Note: The server will only accept write operations if the result fulfils the server Function
926 requirements (permitted values, e.g.). Write operations on Elements that are not writeable MAY
927 result in an error message.

928 M, R or O may be combined with the suffix "(event)" to express that a supported Element or value
929 only has to be supported during a certain event and hence does not need to be present at all times. If
930 the event is not active the Element may be omitted or another value may be set. In most cases a
931 High-Level requirement reference for the event is given in the rules column.

932 In some cases, an Element may have a double presence indication, separated by a comma. Examples:

- 933 - M, O
- 934 - M \W, O \W

The description (rules) of the Element details in such a case the presence requirement. A typical example is a list-based Element where the "first" (or "last") Element has a different presence requirement than the other list Elements.

3.1.3.4 Cardinality indications on Elements and list items

A cardinality indication on an Element or list item expresses constraints on the number of occurrences of a given Element or data set. In this section we use "X" as representation for such an Element or data set. Furthermore, "a" and "b" represent constraints. The following rules apply for the occurrence of "X" and its content related to a specific Scenario (see note underneath the list):

1. X
No cardinality indication.
2. X (a..b)
This means "X" SHALL occur at least "a" times and at maximum "b" times.
3. X (a..
This means "X" SHALL occur at least "a" times and MAY occur more than "a" times.
4. X (..b)
This means "X" SHALL occur at maximum "b" times and MAY occur less than "b" times (even zero occurrences are permissive).

Note: These rules apply only under consideration of presence indications and with regards to the given Scenario or Function definition for this Use Case.

The following table is an example to explain this for two different placements.

Scenario [{...}]: M/R/O [W]\C]	Element	Value	[High-Level Mapping] Element and value rules
...	...	...	...
2: M \W	xFeatureType. xListData. xData. (1..3)		
2: M \W	xId	<*(1..)>	PRIMARY IDENTIFIER
2: M \W	timePeriod		...
2: M \W	timePeriod. startTime	<xs:duration>	
2: M \W	xSlot. (1..)		
2: M \W	xSlot. xSlotId		...
2: M \W	xSlot. duration	<xs:duration>	...
...	...	...	...

Table 7: Example table for cardinality indications on Elements and list items

The field

xFeatureType. xListData. xData. (1..3)

introduces a data pattern (required Elements and values) for "xData" instances used for Scenario 2. The field itself specifies that such an "xData" instance SHALL occur at least 1 time and at maximum 3

times within "xListData" of Feature Type "xFeatureType". However, this constraint holds only for Scenario 2 and only if such "xData" are required. In this case, they are required, as the left field

2: M \W

denotes that this data set is mandatory for Scenario 2.

The field

xSlot. (1..)

expresses that the Element "xSlot" SHALL occur at least one time within its "xData", but MAY occur more than one time.

For the expression "<*#{1..}>" of Element "xId" please see section 3.1.3.6.

The remaining fields do not have an explicit cardinality indication.

Note: Cardinality expressions are also used within placeholder expressions as defined in section 3.1.3.6. In many cases such placeholder expressions define the number of required or permitted Elements or list items as they explicitly define how many different values for a given Identifier are required or permitted for a given Scenario.

3.1.3.5 Writability and changeability indication

In the same column where the presence indications are denoted, a mark is used to distinguish between writeable, changeable or readable Elements:

- Elements that are marked with "\W" are written by a client and SHALL be writeable at the server according to their presence indications. The client is not obliged to read the according data. Received notifications do not need to be evaluated.
- Elements that are marked with "\C" are changed by a client and SHALL be changeable at the server according to their presence indications. The client is not obliged to read the according data. Received notifications do not need to be evaluated.
- Elements that are marked with "\RW" are read and written by a client and SHALL be writeable and provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.
- Elements that are marked with "\RC" are read and changed by a client and SHALL be changeable and provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.
- Elements that are not marked are only read by a client and SHALL be provided by the server according to their presence indications. Received notifications SHALL be evaluated according to their presence indications.

"Writeable" means that the Element and its value may be written by a client. This includes the possibility to modify (if the Element is already present), create (if the Element is not present yet), and delete the Element. The server SHALL adjust its Function according to the received "write" operation (unless the server cannot accept the "write" operation according to section 3.1.3.3).

"Changeable" means that the Element's value may be changed by a client. If the Element is not present at the resource before, it probably **cannot** be created by the client via the "write" operation. In this case the server MAY decline such a message.

Note: "\W" includes "\C" already.

Note: Depending on the resource a client might need to request a proper binding before the server accepts a "write" operation.

3.1.3.6 "Value" placeholders

3.1.3.6.1 Introduction

Specializations may use placeholders to model relations between different Elements or even list items of different Functions. The main purpose is to declare which Identifier values relate to each other. As a Use Case does not prescribe specific values to be used for a given Identifier, a placeholder like "<x1>" can be used in "Value" columns to express the intended relations.

There are two styles to identify a placeholder that can be referenced:

1. <xM>
2. <xM#S>

where

1. "x" is any alphabetical prefix like "m", "t", "ec", "cc", etc.
2. "M" is a (major) number like "1", "2", "15", etc.
3. "S" is a sub-number like "1", "7", "10", etc.

Examples for the first style are "<ec1>", "<z12>". Examples for the second style are "<p4#2>", "<m22#3>". For a given placeholder, only one of the styles can be used.

In addition, there are also styles for placeholders that do not need to be referenced:

1. <*>
2. <*#S>

The second style is only used with so-called cardinality expressions.

3.1.3.6.2 Uniqueness of placeholders

A given placeholder <xM> or <xM#S> represents the same value throughout a given Use Case specification for a given set of its parent Identifier values. This shall be explained in a brief example:

We assume a list item with PRIMARY IDENTIFIER "pId". It also has a SUB IDENTIFIER "sId" with placeholder "<s1>". Then, each occurrence of "<s1>" represents the same value for a given value of pId. This means that "<s1>" of a list item with pId=1 can differ from "<s1>" of a list item with pId=2. But it also means that "<s1>" represents the same value in all list items with pId=1.

1032 Note: Typically, parent Identifiers like "pId" will also be expressed with a placeholder like "<p5>", e.g.
 1033 In this case, the uniqueness rule applies for "<p5>" likewise.

1034 Note: The uniqueness also applies for placeholders used as FOREIGN IDENTIFIER.

1035

1036 3.1.3.6.3 Placeholder expressions with cardinalities

1037 For some Identifiers, more than one placeholder is needed. Several notations are used for this
 1038 purpose, which make use of cardinality expressions. The general notation is as follows:

1039 1. <xM#(a..b)>

1040 This is equivalent to this explicit definition:

1041 At least a and at maximum b placeholders of this list: <xM#1> <xM#2> ... <xM#b>

1042 This means that the implementation of a given Use Case (or Scenario) requires a minimum of "a"
 1043 distinct values of the respective Identifier. In total, there can be up to "b" distinct values of the
 1044 respective Identifier.

1045 Additionally, the following notations may occur:

1046 2. <xM#(a..)>

1047 This is equivalent to "<xM#(a..b)>" with "b" equal to infinity.

1048 3. <xM#(..b)>

1049 This is equivalent to "<xM#(a..b)>" with "a" equal to zero.

1050 "<xM#(a..)>" has only a lower bound of "a" distinct values, but no upper bound. "<xM#(..b)>", on the
 1051 other hand, expresses that the Identifier may not be required at all, but it is permitted to have up to
 1052 "b" distinct values.

1053 Similarly, the cardinality can be used for placeholders that are not referenced, i.e. <*#(a..b)> etc.

1054 Note: The cardinality does NOT express which of the sub-numbers have to be used! I.e., it does NOT
 1055 mean that the Identifiers <xM#1> ... <xM#a> are always used and just those with larger sub-numbers
 1056 (<xM#a+1> ... <xM#b>) are optional. If, for instance, a placeholder expression "<xM#(3..5)>" is given,
 1057 it may well happen that an implementation makes use of <xM#2>, <xM#4>, and <xM#5> (i.e., it does
 1058 NOT use <xM#1>, <xM#3>). Which sub-numbers are used usually depends on other parts of a
 1059 Specialization and their references to required placeholders, which is explained in section 3.1.3.6.4.

1060

1061 3.1.3.6.4 References to placeholders and relations

1062 According to the styles for placeholders that can be referenced, an enumeration value "e" can refer
 1063 to a particular placeholder:

1064 1. e(-><xM>)

1065 2. e(-><xM#S>)

1066 This denotes that "e" is to be used with "<xM>" or "<xM#S>", resp.

1067 Example: A Specialization contains the Elements "mld" and "phase". "mld" is an Identifier with
 1068 placeholder definition <m2#(1..3)>. "phase" is a string that permits the values "a", "b", and "c" using
 1069 this expression:

1070 "a"(-><m2#1>)

1071 "b"(-><m2#2>)

1072 "c"(-><m2#3>)

1073 This expresses that the enumeration value "a" is to be used with the placeholder <m2#1>, "b" is to
 1074 be used with <m2#2> and "c" with <m2#3>.

1075 Similarly, a placeholder "yN" can refer to a particular placeholder:

1076 3. <yN->xM>

1077 4. <yN->xM#S>

1078 5. <yN#T->xM>

1079 6. <yN#T->xM#S>

1080 where "T" is a sub-number of "yN".

1081 It is also feasible to associate placeholders with cardinalities:

1082 7. <yN#(a..b)->xM#(a..b)>

1083 denotes that <yN#1> is to be used with <xM#1>, <yN#2> is to be used with <xM#2>, etc.

1084 Note: In this case, the placeholder expressions of yN and xM must have the same cardinality.

1085 In some cases, there is a need to express that multiple list items with similar values are feasible or
 1086 required, but only particular combinations of these different data are permitted. The following
 1087 example shows that several "fData" list items with different "phase" value are required, but that
 1088 these list items may only express either the "phase" value combination { "a", "b", "c" } or the "phase"
 1089 value combination { "a", "abc", "neutral" }. The permitted combinations are defined in a note below a
 1090 table:

Scenario {...}: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	F. fListData. fData.		
2: M	zld	<z3#(3..5)>	
2: M	phase	"a"(-><z3#1>)	
		"b"(-><z3#2>)	
		"c"(-><z3#3>)	
		"abc"(-><z3#4>)	
		"neutral"(-><z3#5>)	

1091 Table 8: Content of an example Specialization

1092 Note: One of the following combinations SHALL be used at least: {<z3#1>, <z3#2>, <z3#3>} or
1093 {<z3#1>, <z3#4>, <z3#5>}.
1094

1095 **3.1.3.7 Rules for content of "Value" column**

1096 For a given Scenario, the "Value" column may restrict the permitted content of a Function's Element
1097 to one or more particular values. This means that Elements with values deviating from the restriction
1098 (i.e. from the permitted values) do not belong to the respective Scenario and need to be considered
1099 as if the Element is not set. If more than one particular value is permitted for an Element, the values
1100 are in a single line each.

1101 If a presence indication is set for the value (in an additional column before the value), the following
1102 rules SHALL be applied:

- 1103 - "M" means that the value SHALL be supported. This means the value needs to be set at a
1104 certain point in time (depending on the value rules) or for a certain Element within a list
1105 entry.
- 1106 - "R" means that the value SHOULD be supported.
- 1107 - "O" means that the value MAY be supported.

1108 If all possible values of a given mandatory Element are optional or recommended and this Element is
1109 used for the purpose of the respective Scenario, one of the values SHALL be set. If all possible values
1110 of a given optional or recommended Element are optional or recommended, this Element MAY
1111 contain also other values, but then this Element SHALL NOT be considered as part of the respective
1112 Scenario.

1113 M, R or O may be combined with the suffix "(event)" to express that a supported value only has to be
1114 supported during a certain event and hence does not need to be present at all times. If the event is
1115 not active another value may be set. In most cases a High-Level requirement reference for the event
1116 is given in the rules column.

1117 If no presence indication is set for the value, the following rules SHALL be applied:

- 1118 - In case of Elements where the server may set or change an Element on its own (see section
1119 3.1.3.5):
 - 1120 ○ within the tables in the "Server data - Resources" sections:
 - 1121 ■ the server SHALL support at least one of the listed values.
 - 1122 ○ within the tables in the "Client data - Specializations" sections:
 - 1123 ■ the client SHALL support all listed values.
- 1124 - In case of Elements that are writeable or changeable (see section 3.1.3.5):
 - 1125 ○ within the tables in the "Server data - Resources" sections:
 - 1126 ■ the server SHALL support all listed values.
 - 1127 ○ within the tables in the "Client data - Specializations" sections:
 - 1128 ■ the client SHALL support at least one of the listed values.

1129 Depending on the Element, different values may be used during runtime. If this is the case, those
1130 rules are described within the value rules.

1131 If a value is placed in parenthesis, the corresponding value is a recommendation. The actual value
 1132 MAY deviate from this, e.g. "(1024)".

1133

1134 **3.1.3.8 General information on how to interpret the "Content of Function..." and "Content of** 1135 **Specialization..." tables**

1136 Within the "Client data - Specializations" sections each Specialization is described in an own sub-
 1137 section with the name "Specialization "<name of the Specialization>" (e.g. "Specialization
 1138 "Measurement_GridFeedInEnergy"). It contains only one table that includes all Elements needed for
 1139 this Specialization. The different Functions are mentioned in a continuous row, highlighted with grey
 1140 background colour. This row contains the following parts:

1141 <Feature Type>. <Function>.[<list entry instance name>.]

1142 The <list entry instance name> is only included if the <Function> is a list-based Function. An example
 1143 could be:

1144 DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData.
 1145 deviceConfigurationKeyValueDescriptionData.

1146 In the following rows, only the names of the Elements are stated, without the prefix described above.

1147

1148 Within the "Server data - Resources" sections each Feature Type is described in an own sub-section
 1149 with the name "Feature Type "<name of the Feature Type>" (e.g. "Feature Type "Measurement").
 1150 It contains sub-sections for each Function named "Function "<name of the Function>" (e.g.
 1151 "Function "measurementListData"). These sections contain one table with all Elements needed for
 1152 this resource. The list entries are mentioned in a continuous row, highlighted with grey background
 1153 colour. This row contains the following parts:

1154 <Feature Type>. <Function>.[<list entry instance name>.]

1155 The <list entry instance name> is only included if the <Function> is a list-based Function. An example
 1156 could be:

1157 Measurement. measurementDescriptionListData. measurementDescriptionData.

1158 In the following rows, only the names of the Elements are stated, without the prefix described above.

1159

1160 For both kinds of tables, the following applies:

1161 - Parent Elements are marked with a dot at the end of the name:

1162 <parent Element>.

1163 E.g.:

1164 value.

1165 - If there are sub-Elements, they are described in own rows with the name of the parent
 1166 Element as prefix, separated by a dot and a blank space:

1167 <parent Element>. <sub-Element>

1168 E.g.:

1169 value. number

1170

1171 3.1.4 Rules for "Feature Types and Functions..." tables

1172 3.1.4.1 Presence indications for "Feature Types and Functions..." tables

1173 The following presence indications are used:

Abbreviation	Meaning	Link to requirement keywords
M	Mandatory	SHALL
R	Recommended	SHOULD
O	Optional	MAY

1174 Table 9: Presence indication of Feature Types and Functions support

1175 If at least one Function of a Feature has the presence indication "M", it is mandatory to support the
1176 Feature.

1177

1178 3.1.4.2 Rules for "Possible operations" column

1179 Within the "Feature Types and Functions..." tables the column "Possible operations" state whether
1180 the Function is read- or writeable (as defined in the detailed discovery mechanism, see
1181 [ProtocolSpecification]).

1182 If the "partial" concept (also called "restricted function exchange") SHALL be supported, the
1183 following notation is used (separated for read and write access):

1184 read (M). partial (M)

1185 write (M). partial (M)

1186 If the "partial" concept SHOULD be supported, the following notation is used:

1187 read (M). partial (R)

1188 write (M). partial (R)

1189 If the "partial" concept MAY be supported, the following notation is used:

1190 read (M). partial (O)

1191 write (M). partial (O)

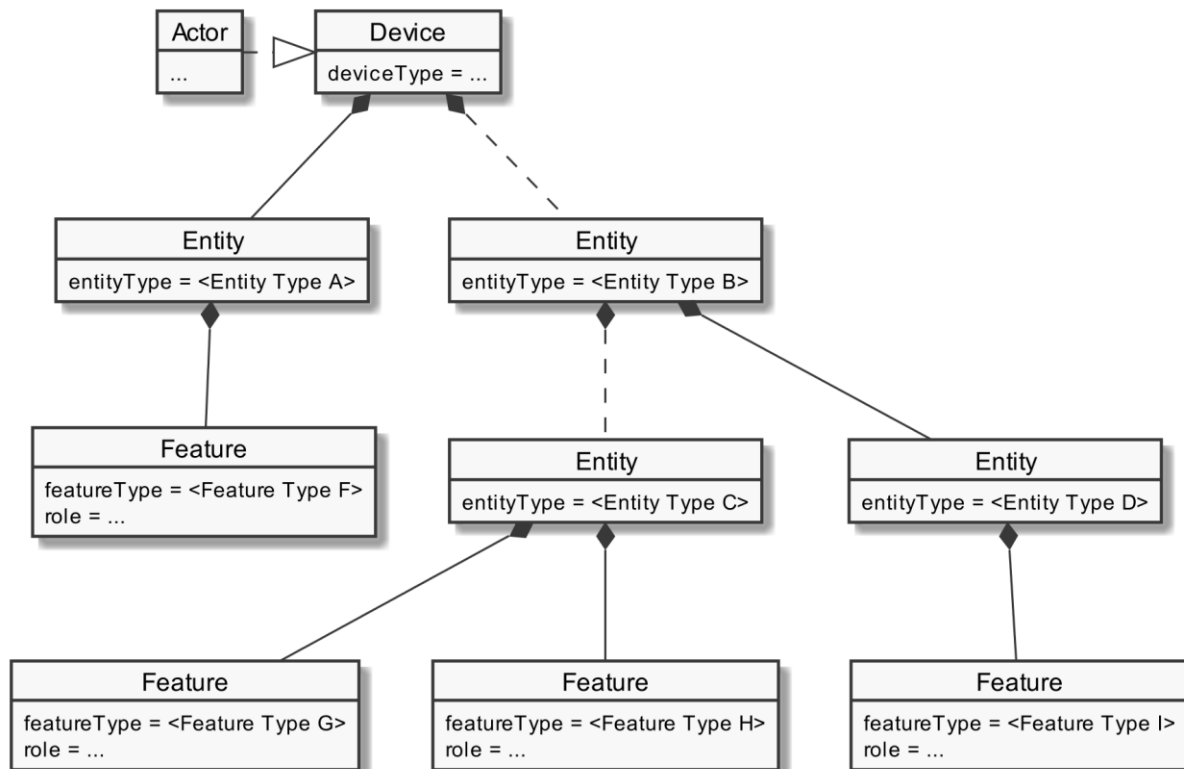
1192 The server can decide whether a notification is submitted complete or partial (as described in
1193 [ProtocolSpecification]) if not defined differently within this Use Case Specification.

1194

1195 3.1.5 "Actor ... overview" diagram rules

1196 Within the "Actor [...] overview" diagrams in the "Actors" sub-sections the complete functionality of
1197 this Use Case is provided, including optional Scenarios. Which Scenarios are optional can be found in
1198 Table 2. The Actor MAY have more functionality implemented than needed for this Use Case.

1199 For the following Actor overview example, a brief description of the graphical symbols will be
 1200 described.



1201
 1202 *Figure 7: Actor overview example*

1203 The solid lines in the figure represent an immediate parent-childhood relation: The Entity with
 1204 "<Entity Type A>" is a direct child of "Device". The Entity with "<Entity Type D>" is a direct child of the
 1205 Entity with "<Entity Type B>". All Features are immediate child of the respective Entity.

1206 The dashed lines in the figure express that there MAY be additional Entities between the shown
 1207 Entities: A vendor's implementation MAY have one or more Entities between "Device" and the Entity
 1208 with "<Entity Type B>". Likewise, a vendor's implementation MAY have one or more Entities between
 1209 the Entity with "<Entity Type B>" and the Entity with "<Entity Type C>".

1211 3.1.6 Specializations

1212 Within the "Actors" sub-sections, Specializations are referenced. A Specialization describes a dataset
 1213 necessary to fulfil the specific requirements of a High-Level Use Case and its Scenarios. Often, data
 1214 from multiple different Features and Functions are needed to fulfil the requirements. Therefore, a
 1215 Specialization defines a dataset that may encompass multiple related Functions from one or more
 1216 different Features.

1217 As different Use Cases sometimes share similar requirements, Specializations are also important
 1218 from a re-usability perspective. This approach is used to improve consistency across Use Cases and
 1219 avoid multiple variances of basically the same dataset. This is especially important in the case when
 1220 an implementation supports multiple Use Cases. E.g. if a power measurement is necessary in two
 1221 different Use Cases, both Use Cases could define slightly different datasets. In this case, the server as

well as the client functionality would have to implement both variances if both Use Cases are supported. This means, depending on the number of Use Cases, two or more datasets need to be generated, transmitted and stored instead of one. Therefore, common Specializations are defined to enable consolidated implementations. Specializations are detailed per Actor, as distinct Actors may use different parts of a Specialization or even have different permissions to change data. In general, this may also depend on the respective Scenario.

If a Feature server can provide the data of a Specialization, the data does not necessarily always need to be available at the Feature server. There might be situations where the user deactivates a Use Case. There may also be other reasons why Use Case data cannot be provided currently. Therefore, a client always needs to be subscribed (as described in section 3.3.4) on the corresponding dataset to stay updated.

The SPINE resource descriptions given in the "SPINE resources of the Actor" sections are derived from the Specializations given in the Actor's overview diagram.

3.1.7 Order of messages within the sequence diagrams

There are several sequence diagrams in this document describing message flows. The order of the messages SHOULD be kept by the communications partners, but there might be cases where a different order makes sense. The communications partners SHALL be able to handle the Scenario functionalities even if the messages are transmitted in a different order by the other Actor(s). The sequence diagrams can be seen as examples.

3.1.8 Further information and rules

3.1.8.1 Frequently used Element rules for the Resource and Specialization tables

This section serves as a collection of rules frequently used by Resource and Specialization tables of the subsequent sections. Each rule applies only where referenced explicitly in the tables.

Note: The purpose of this collection is just to reduce the size of the tables. As such, no rule has a meaning without a reference indicating the required rule. A reference looks like "See [Scaled number rules]", e.g.

[Scaled number rules]:

The sub-Elements "number" and "scale" represent a value according to the following formula:
$$\text{number} * 10^{\text{scale}}$$

3.1.8.2 Rules regarding the usage of time-related information

Timestamps used within this Use Case SHALL be presented as absolute times. This means relative times SHALL NOT be used for timestamps.

1258 It needs to be distinguished between timestamps with date information and without date
1259 information:

- 1260 - For [POEN-011], [POEN-012], [POEN-021] and [POEN-022], absolute timestamps with date
1261 information SHALL be used ([POEN-001/1]).
- 1262 - For [POEN-031] and [POEN-032], absolute timestamps without date information SHALL be
1263 used ([POEN-001/2]).

1264 Durations used within this Use Case SHALL be presented as relative times ([POEN-001/3]).

1265 Note: This means the Actors need to have a common understanding of absolute times. It is strongly
1266 recommended that each Actor is "sufficiently synchronised" with the official "Coordinated Universal
1267 Time" (UTC).

1268

1269 **3.1.8.3 Applied sign convention**
1270 For the applied sign convention, please see section 2.6.1.

1271

1272 **3.2 Actors**

1273 **3.2.1 Transmission Broker**

1274 **3.2.1.1 Resource hierarchy**
1275 If Use Case discovery is supported (see section 3.1.2) this Actor SHALL be denoted as
1276 "TransmissionBroker" in the Element "nodeManagementUseCaseData. useCaseInformation. actor".

1277 The following diagram provides an overview of the Actor Transmission Broker's resource hierarchy.

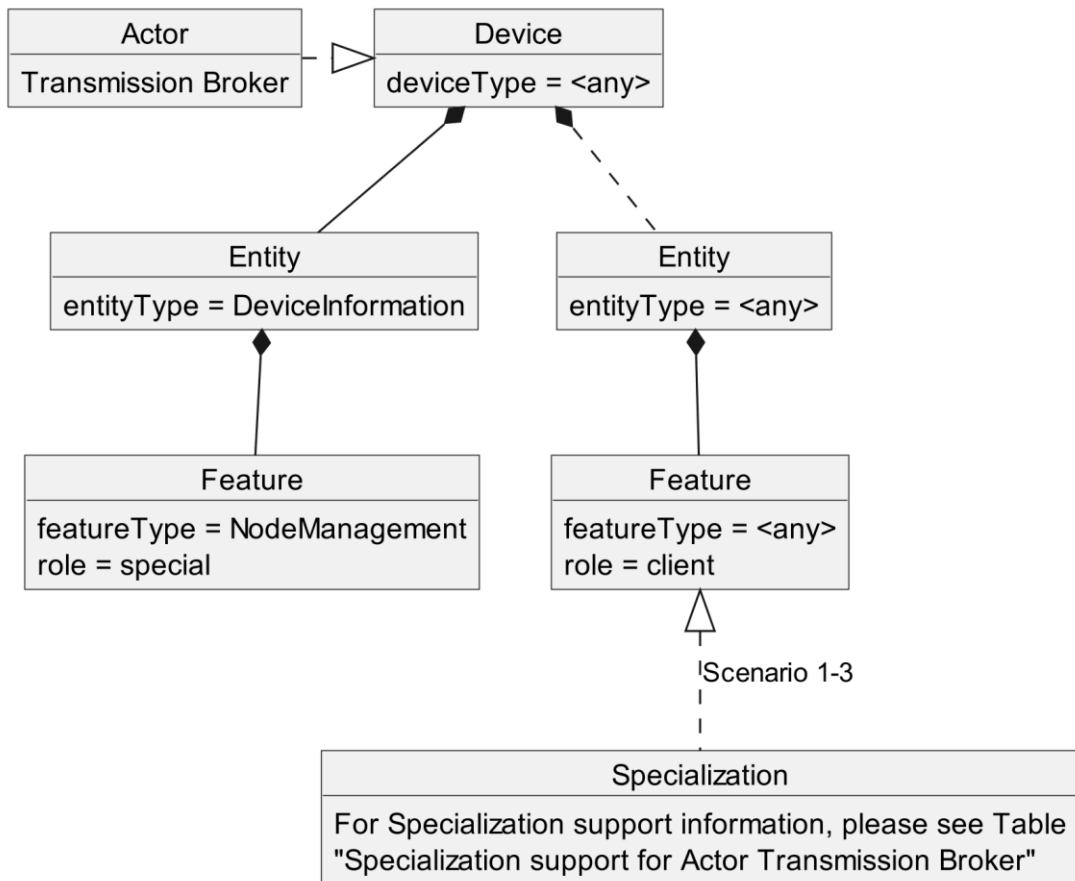


Figure 8: Actor "Transmission Broker" overview

The "Actor ... overview" diagram rules" section describes how to interpret the diagram above. See the "Specializations" section for more information regarding the Specializations given in the diagram above.

Note: The entityType "DeviceInformation" with the featureType "NodeManagement" is required by the SPINE protocol and therefore SHALL be supported. Both types are added in the figure for completeness but are not directly linked to the Use Case.

The Use Case specific data follow behind any entityType. The Specializations represent the Scenario specific data that must be supported for each Scenario and are realized through the corresponding featureTypes.

If a Specialization is connected to a Feature with the role "client", the Actor has a client role for this data. This means that the Actor accesses the data set described by the Specialization at a corresponding server Feature. Further details are described in the sub-section "Client data - Specializations".

If a Specialization is connected to a Feature with the role "server", the Actor has the server role for this data. This means that the Actor must provide the corresponding data set of the Specialization as part of its Features. Further details are described in the sub-section "Server data - Resources".

Specialization name	Scenario	Described in table
TimeSeries_ActivePowerConsumptionLimitCurve	1	Table 11
TimeSeries_ActivePowerProductionLimitCurve	2	Table 12
TimeSeries_FallbackActivePowerConsumptionLimitCurve	3	Table 13
TimeSeries_FallbackActivePowerProductionLimitCurve	3	Table 14

Table 10: Specialization support for Actor Transmission Broker

3.2.1.2 Server data - Resources

As this Actor has only client functionality, no resources are described within this section.

3.2.1.3 Client data - Specializations

3.2.1.3.1 Topic "TimeSeries"

3.2.1.3.1.1 Specialization "TimeSeries_ActivePowerConsumptionLimitCurve"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	TimeSeries.timeSeriesDescriptionListData.timeSeriesDescriptionData.		
1: M	timeSeriesId	<ts1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: M	timeSeriesType	"consumptionLimitCurve"	"consumptionLimitCurve" SHALL be used to model a consumption limit over time.
1: M	timeSeriesWriteable	"true"	If set to "true", the corresponding timeSeriesData SHALL be writeable.
1: O	measurementId	<m*#(1..1)>	FOREIGN IDENTIFIER. If measurementId is used, it SHALL link to a measurement value of the total active power.
1: M	unit	"W"	The unit SHALL be applied to the value in timeSeriesData
1: M	scopeType	"activePowerLimit"	
1: M	TimeSeries.timeSeriesConstraintsListData.timeSeriesConstraintsData		
1: M	timeSeriesId	<ts1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: O	slotCountMin		[POEN-013/1] If set, the corresponding timeSeriesData SHALL NOT have less slots.
1: M	slotCountMax	≥192	[POEN-013/2] If set, the corresponding timeSeriesData SHALL NOT have more slots.
1: O	slotDurationMin		[POEN-001/3], [POEN-013/3]

			If set, the corresponding timeSeriesData SHALL NOT have slots with shorter duration.
1: O	slotDurationMax		[POEN-001/3], [POEN-013/4] If set, the corresponding timeSeriesData SHALL NOT have slots with longer duration.
1: O	slotDurationStepSize		[POEN-013/5] If set, the duration of a slot SHALL be a multiple of slotDurationStepSize.
1: M	slotValueMin.* ¹		[POEN-002/1], [POEN-013/6] [Scaled number rules] The Elements "value", "maxValue" and "minValue" of the corresponding timeSeriesData's slots SHALL be equal to or higher than slotValueMin.
1: M	slotValueMin. number	0	SHALL be used.
1: M	slotValueMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
1: M	slotValueMax.* ¹		[POEN-002/1], [POEN-013/7] [Scaled number rules] The Elements "value", "maxValue" and "minValue" of the corresponding timeSeriesData's slots SHALL be equal to or lower than slotValueMax.
1: M	slotValueMax. number	≥0	SHALL be used.
1: M	slotValueMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
1: O	slotValueStepSize.		[POEN-002/2], [POEN-013/8] [Scaled number rules] The Elements "value", "maxValue" and "minValue" of the corresponding timeSeriesData's slots SHALL be a multiple of slotValueStepSize.
1: M	slotValueStepSize. number	≥0	SHALL be used.
1: M	slotValueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
1: M	TimeSeries. timeSeriesListData. timeSeriesData.		
1: M	timeSeriesId	<ts1#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
1: F	timePeriod.		
1: M \RW	timeSeriesSlot.		
1: M \RW	timeSeriesSlot. timeSeriesSlotId		
1: M \RW	timeSeriesSlot. timePeriod.		
1: M \RW	timeSeriesSlot. timePeriod. startTime	<xs:dateTime>	[POEN-001/1]

1: M \RW	timeSeriesSlot. duration	>0	[POEN-001/3], [POEN-005/1], [POEN-005/2] SHALL be set and express the duration of the slot. SHALL only contain values greater than zero seconds. The duration SHALL last to the start time of the next slot, if any.
1: M \RW	timeSeriesSlot. minValue.* ²		[POEN-002/1], [POEN-006], [POEN-011] [Scaled number rules]
1: M \RW	timeSeriesSlot. minValue. number	≥0	SHALL be used.
1: M \RW	timeSeriesSlot. minValue. scale		SHALL be used.
1: M \RW	timeSeriesSlot. maxValue.* ²		[POEN-002/1], [POEN-006], [POEN-012] [Scaled number rules]
1: M \RW	timeSeriesSlot. maxValue. number	≥0	SHALL be used.
1: M \RW	timeSeriesSlot. maxValue. scale		SHALL be used.

Table 11: Content of Specialization "TimeSeries_ActivePowerConsumptionLimitCurve" at Actor Transmission Broker

*¹: The Elements slotValueMin and slotValueMax SHALL be interpreted mathematically.

*²: The Elements minValue and maxValue SHALL be interpreted logically (which corresponds to the mathematical way for the consumption direction).

3.2.1.3.1.2 Specialization "TimeSeries_ActivePowerProductionLimitCurve"

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
2: M	TimeSeries.timeSeriesDescriptionListData.timeSeriesDescriptionData.		
2: M	timeSeriesId	<ts2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: M	timeSeriesType	"productionLimitCurve"	"productionLimitCurve" SHALL be used to model a production limit over time.
2: M	timeSeriesWriteable	"true"	If set to "true", the corresponding timeSeriesData SHALL be writeable.
2: O	measurementId	<m*#(1..1)>	FOREIGN IDENTIFIER. If measurementId is used, it SHALL link to a measurement value of the total active power.
2: M	unit	"W"	The unit SHALL be applied to the value in timeSeriesData
2: M	scopeType	"activePowerLimit"	
2: M	TimeSeries.timeSeriesConstraintsListData.timeSeriesConstraintsData		

2: M	timeSeriesId	<ts2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
2: O	slotCountMin		[POEN-023/1] If set, the corresponding timeSeriesData SHALL NOT have less slots.
2: M	slotCountMax	≥192	[POEN-023/2] If set, the corresponding timeSeriesData SHALL NOT have more slots.
2: O	slotDurationMin		[POEN-001/3], [POEN-023/3] If set, the corresponding timeSeriesData SHALL NOT have slots with shorter duration.
2: O	slotDurationMax		[POEN-001/3], [POEN-023/4] If set, the corresponding timeSeriesData SHALL NOT have slots with longer duration.
2: O	slotDurationStepSize		[POEN-023/5] If set, the duration of a slot SHALL be a multiple of slotDurationStepSize.
2: M	slotValueMin.* ¹		[POEN-002/1], [POEN-023/6] [Scaled number rules] The Elements "value", "maxValue" and "minValue" of the corresponding timeSeriesData's slots SHALL be equal to or higher than slotValueMin.
2: M	slotValueMin. number	≤0	SHALL be used.
2: M	slotValueMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: M	slotValueMax.* ¹		[POEN-002/1], [POEN-023/7] [Scaled number rules] The Elements "value", "maxValue" and "minValue" of the corresponding timeSeriesData's slots SHALL be equal to or lower than slotValueMax.
2: M	slotValueMax. number	0	SHALL be used.
2: M	slotValueMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: O	slotValueStepSize.		[POEN-002/2], [POEN-023/8] [Scaled number rules] The Elements "value", "maxValue" and "minValue" of the corresponding timeSeriesData's slots SHALL be a multiple of slotValueStepSize.
2: M	slotValueStepSize. number	≥0	SHALL be used.
2: M	slotValueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
2: M	TimeSeries. timeSeriesListData. timeSeriesData.		
2: M	timeSeriesId	<ts2#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.

2: F	timePeriod.		
2: M \RW	timeSeriesSlot.		
2: M \RW	timeSeriesSlot. timeSeriesSlotId		
2: M \RW	timeSeriesSlot. timePeriod.		
2: M \RW	timeSeriesSlot. timePeriod. startTime	<xs:dateTime>	[POEN-001/1]
2: M \RW	timeSeriesSlot. duration	>0	[POEN-001/3], [POEN-005/1], [POEN-005/2] SHALL be set and express the duration of the slot. SHALL only contain values greater than zero seconds. The duration SHALL last to the start time of the next slot, if any.
2: M \RW	timeSeriesSlot. minValue.* ²		[POEN-002/1], [POEN-006], [POEN-021] [Scaled number rules]
2: M \RW	timeSeriesSlot. minValue. number	≤0	SHALL be used.
2: M \RW	timeSeriesSlot. minValue. scale		SHALL be used.
2: M \RW	timeSeriesSlot. maxValue.* ²		[POEN-002/1], [POEN-006], [POEN-022] [Scaled number rules]
2: M \RW	timeSeriesSlot. maxValue. number	≤0	SHALL be used.
2: M \RW	timeSeriesSlot. maxValue. scale		SHALL be used.

Table 12: Content of Specialization "TimeSeries_ActivePowerProductionLimitCurve" at Actor Transmission Broker

*¹: The Elements slotValueMin and slotValueMax SHALL be interpreted mathematically.

*²: The Elements minValue and maxValue SHALL be interpreted logically (not mathematically). This means that the minValue is mathematically higher than the maxValue.

3.2.1.3.1.3 Specialization "TimeSeries_FallbackActivePowerConsumptionLimitCurve"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
3: M	TimeSeries. timeSeriesDescriptionListData. timeSeriesDescriptionData.		
3: M	timeSeriesId	<ts3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: M	timeSeriesType	"fallbackConsumption LimitCurve"	"fallbackConsumptionLimitCurve" SHALL be used to model the fallback consumption limit curve.

3: M	timeSeriesWriteable	"true"	If set to "true", the corresponding timeSeriesData SHALL be writeable.
3: O	measurementId	<m*#(1..1)>	FOREIGN IDENTIFIER. If measurementId is used, it SHALL link to a measurement value of the total active power.
3: M	unit	"W"	The unit SHALL be applied to the value in timeSeriesData
3: M	scopeType	"activePowerLimit"	
3: M	TimeSeries. timeSeriesConstraintsListData. timeSeriesConstraintsData		
3: M	timeSeriesId	<ts3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: O	slotCountMin		[POEN-033/1] If set, the corresponding timeSeriesData SHALL NOT have less slots.
3: M	slotCountMax	≥96	[POEN-033/2] If set, the corresponding timeSeriesData SHALL NOT have more slots.
3: O	slotDurationMin		[POEN-001/3], [POEN-033/3] If set, the corresponding timeSeriesData SHALL NOT have slots with shorter duration.
3: O	slotDurationMax		[POEN-001/3], [POEN-033/4] If set, the corresponding timeSeriesData SHALL NOT have slots with longer duration.
3: O	slotDurationStepSize		[POEN-033/5] If set, the duration of a slot SHALL be a multiple of slotDurationStepSize.
3: M	slotValueMin.* ¹		[POEN-002/1], [POEN-033/6] [Scaled number rules] The Elements "value", "maxValue" and "minValue" of the corresponding timeSeriesData's slots SHALL be equal to or higher than slotValueMin.
3: M	slotValueMin. number	0	SHALL be used.
3: M	slotValueMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: M	slotValueMax.* ¹		[POEN-002/1], [POEN-033/7] [Scaled number rules] The Elements "value", "maxValue" and "minValue" of the corresponding timeSeriesData's slots SHALL be equal to or lower than slotValueMax.
3: M	slotValueMax. number	≥0	SHALL be used.
3: M	slotValueMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: O	slotValueStepSize.		[POEN-002/2], [POEN-033/8] [Scaled number rules]

			The Elements "value", "maxValue" and "minValue" of the corresponding timeSeriesData's slots SHALL be a multiple of slotValueStepSize.
3: M	slotValueStepSize. number	≥0	SHALL be used.
3: M	slotValueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: M	TimeSeries. timeSeriesListData. timeSeriesData.		
3: M	timeSeriesId	<ts3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: F	timePeriod.		
3: M \RW	timeSeriesSlot.		[POEN-009]
3: M \RW	timeSeriesSlot. timeSeriesSlotId		
3: M \RW	timeSeriesSlot. timePeriod.		
3: M \RW	timeSeriesSlot. timePeriod. startTime	<xs:time>	[POEN-001/1], [POEN-001/2]
3: M \RW	timeSeriesSlot. duration	>0	[POEN-001/3], [POEN-005/1], [POEN-005/2] SHALL be set and express the duration of the slot. SHALL only contain values greater than zero seconds. The duration SHALL last to the start time of the next slot, if any.
3: M \RW	timeSeriesSlot. maxValue.* ²		[POEN-002/1], [POEN-006], [POEN-007], [POEN-008], [POEN-031] [Scaled number rules]
3: M \RW	timeSeriesSlot. maxValue. number	≥0	SHALL be used.
3: M \RW	timeSeriesSlot. maxValue. scale		SHALL be used.

Table 13: Content of Specialization "TimeSeries_FallbackActivePowerConsumptionLimitCurve" at Actor Transmission Broker

*¹: The Elements slotValueMin and slotValueMax SHALL be interpreted mathematically.

*²: The Element maxValue SHALL be interpreted logically (which corresponds to the mathematical way for the consumption direction).

3.2.1.3.1.4 Specialization "TimeSeries_FallbackActivePowerProductionLimitCurve"

Scenario [{...}]: M/R/O [\W][\C]	Element	Value	[High-Level Mapping] Element and value rules
3: M	TimeSeries. timeSeriesDescriptionListData. timeSeriesDescriptionData.		
3: M	timeSeriesId	<ts4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.

3: M	timeSeriesType	"fallbackProductionLimitCurve"	"fallbackProductionLimitCurve" SHALL be used to model the fallback peoduction limit curve.
3: M	timeSeriesWriteable	"true"	If set to "true", the corresponding timeSeriesData SHALL be writeable.
3: O	measurementId	<m*#(1..1)>	FOREIGN IDENTIFIER. If measurementId is used, it SHALL link to a measurement value of the total active power.
3: M	unit	"W"	The unit SHALL be applied to the value in timeSeriesData
3: M	scopeType	"activePowerLimit"	
3: M	TimeSeries. timeSeriesConstraintsListData. timeSeriesConstraintsData		
3: M	timeSeriesId	<ts4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: O	slotCountMin		[POEN-034/1] If set, the corresponding timeSeriesData SHALL NOT have less slots.
3: M	slotCountMax	≥96	[POEN-034/2] If set, the corresponding timeSeriesData SHALL NOT have more slots.
3: O	slotDurationMin		[POEN-001/3], [POEN-034/3] If set, the corresponding timeSeriesData SHALL NOT have slots with shorter duration.
3: O	slotDurationMax		[POEN-001/3], [POEN-034/4] If set, the corresponding timeSeriesData SHALL NOT have slots with longer duration.
3: O	slotDurationStepSize		[POEN-034/5] If set, the duration of a slot SHALL be a multiple of slotDurationStepSize.
3: M	slotValueMin.* ¹		[POEN-002/1], [POEN-034/6] [Scaled number rules] The Elements "value", "maxValue" and "minValue" of the corresponding timeSeriesData's slots SHALL be equal to or higher than slotValueMin.
3: M	slotValueMin. number	≤0	SHALL be used.

3: M	slotValueMin. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: M	slotValueMax.* ¹		[POEN-002/1], [POEN-034/7] [Scaled number rules] The Elements "value", "maxValue" and "minValue" of the corresponding timeSeriesData's slots SHALL be equal to or lower than slotValueMax.
3: M	slotValueMax. number	0	SHALL be used.
3: M	slotValueMax. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: O	slotValueStepSize.		[POEN-002/2], [POEN-034/8] [Scaled number rules] The Elements "value", "maxValue" and "minValue" of the corresponding timeSeriesData's slots SHALL be a multiple of slotValueStepSize.
3: M	slotValueStepSize. number	≥0	SHALL be used.
3: M	slotValueStepSize. scale		SHALL be interpreted. If absent, a default value of "0" applies.
3: M	TimeSeries. timeSeriesListData. timeSeriesData.		
3: M	timeSeriesId	<ts4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: F	timePeriod.		
3: M \RW	timeSeriesSlot.		[POEN-009]
3: M \RW	timeSeriesSlot. timeSeriesSlotId		
3: M \RW	timeSeriesSlot. timePeriod.		
3: M \RW	timeSeriesSlot. timePeriod. startTime	<xs:time>	[POEN-001/1], [POEN-001/2]
3: M \RW	timeSeriesSlot. duration	>0	[POEN-001/3], [POEN-005/1], [POEN-005/2] SHALL be set and express the duration of the slot. SHALL only contain values greater than zero seconds. The duration SHALL last to the start time of the next slot, if any.

3: M \RW	timeSeriesSlot. maxValue.* ²		[POEN-002/1], [POEN-006], [POEN-007], [POEN-008], [POEN-032] [Scaled number rules]
3: M \RW	timeSeriesSlot. maxValue. number	≤0	SHALL be used.
3: M \RW	timeSeriesSlot. maxValue. scale		SHALL be used.

Table 14: Content of Specialization "TimeSeries_FallbackActivePowerProductionLimitCurve" at Actor Transmission Broker

*¹: The Elements slotValueMin and slotValueMax SHALL be interpreted mathematically.

*²: The Element maxValue SHALL be interpreted logically (not mathematically). This means that the maxValue is the mathematically smallest value that is permitted for the power production (as production power is provided as negative values).

3.2.2 CEM

3.2.2.1 Resource hierarchy

If Use Case discovery is supported (see section 3.1.2) this Actor SHALL be denoted as "CEM" in the Element "nodeManagementUseCaseData. useCaseInformation. actor".

The following diagram provides an overview of the Actor CEM's resource hierarchy.

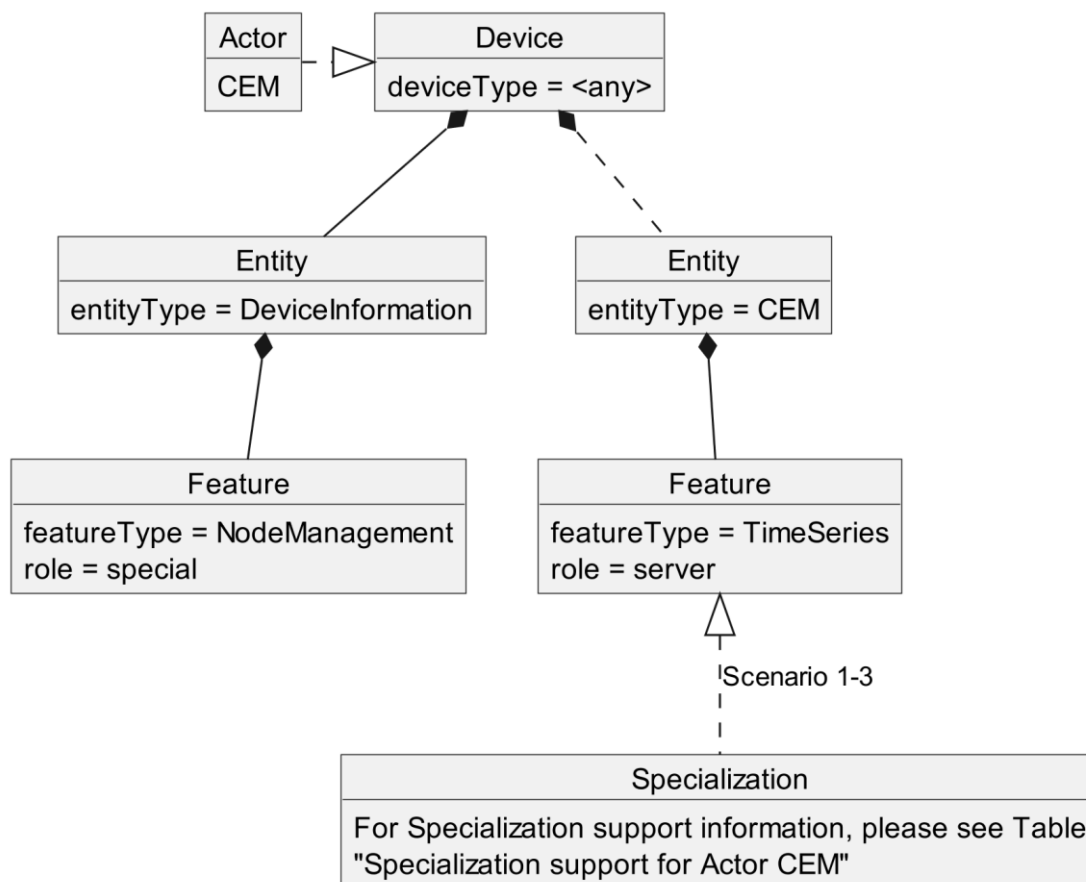


Figure 9: Actor "CEM" overview

The ""Actor ... overview" diagram rules" section describes how to interpret the diagram above. See the "Specializations" section for more information regarding the Specializations given in the diagram above.

Note: The entityType "DeviceInformation" with the featureType "NodeManagement" is required by the SPINE protocol and therefore SHALL be supported. Both types are added in the figure for completeness but are not directly linked to the Use Case.

The Use Case specific data follow behind the entityType "CEM". The Specializations represent the Scenario specific data that must be supported for each Scenario and are realized through the corresponding featureTypes.

If a Specialization is connected to a Feature with the role "client", the Actor has a client role for this data. This means that the Actor accesses the data set described by the Specialization at a corresponding server Feature. Further details are described in the sub-section "Client data - Specializations".

If a Specialization is connected to a Feature with the role "server", the Actor has the server role for this data. This means that the Actor must provide the corresponding data set of the Specialization as part of its Features. Further details are described in the sub-section "Server data - Resources".

Specialization name	Scenario	Used Feature...	...in tables
TimeSeries_ActivePowerConsumptionLimitCurve	1	TimeSeries	Table 17
			Table 18
			Table 19
TimeSeries_ActivePowerProductionLimitCurve	2	TimeSeries	Table 17
			Table 18
			Table 19
TimeSeries_FallbackActivePowerProductionLimitCurve	3	TimeSeries	Table 17
			Table 18
			Table 19
TimeSeries_FallbackActivePowerProductionLimitCurve	3	TimeSeries	Table 17
			Table 18
			Table 19

Table 15: Specialization support for Actor CEM

3.2.2.2 Server data - Resources

3.2.2.2.1 Overview

Behind the entityType "CEM", the Actor CEM SHALL offer the Feature Types and Functions given in the table below.

Feature Type	Scenario: M/R/O	Function	Possible operations
TimeSeries	1: M 2: M 3: M	timeSeriesDescriptionListData	read (M). partial (R)

	1: M 2: M 3: M	timeSeriesConstraintsListData	read (M). partial (R)
	1: M 2: M 3: M	timeSeriesListData	read (M). partial (R) write (M). partial (M)

1357 Table 16: Feature Types and Functions used within this Use Case by the Actor CEM

1358 For each of these Feature Types the following rule applies: There SHALL be at maximum one Feature
1359 with the Feature Type in the Entity.

1360 Note: As a consequence of the previous rule, an implementation may need to have Feature data
1361 from different Scenarios/Specializations or even Use Cases in a given Feature.

1362 The Scenario number shows in which Scenarios the CEM acts as server and which Feature Types and
1363 Functions are relevant in each Scenario.

1364 A detailed definition of the Elements and values that shall be supported in each Function is given in
1365 the following sub-sections.

1366 Note: If in the table above "partial" read is not mentioned or is only optional, it still might be
1367 mandatory to support partial notifications. The details of "partial" support are described within the
1368 Scenario sections.

1369 Note: The presence indications stated above are meant relative to the ones of the according Scenario
1370 stated in Table 2. I.e. if a Scenario is optional ("O") and a Feature Type is mandatory ("M") the
1371 Feature Type need only be supported if the Scenario is supported, too.

1372 Note: Further Features MAY be implemented on the same Entities, as well as further Functions MAY
1373 be implemented in the used Entities.

1374

1375 3.2.2.2.2 Feature Type "TimeSeries"

1376 3.2.2.2.2.1 Function "timeSeriesDescriptionListData"

Scenario [...]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	TimeSeries. timeSeriesDescriptionListData. timeSeriesDescriptionData.		
1: M	timeSeriesId	<ts1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
1: M	timeSeriesType	"consumptionLimitCurve"	"consumptionLimitCurve" SHALL be used to model a consumption limit over time.
1: M	timeSeriesWriteable	"true"	If set to "true", the corresponding

			timeSeriesData SHALL be writeable.
1: O	measurementId	<m*#{1..1}>	FOREIGN IDENTIFIER. If measurementId is used, it SHALL link to a measurement value of the total active power.
1: M	unit	"W"	The unit SHALL be applied to the value in timeSeriesData
1: M	scopeType	"activePowerLimit"	
2: M	TimeSeries. timeSeriesDescriptionListData. timeSeriesDescriptionData.		
2: M	timeSeriesId	<ts2#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: M	timeSeriesType	"productionLimitCurve"	"productionLimitCurve" SHALL be used to model a production limit over time.
2: M	timeSeriesWriteable	"true"	If set to "true", the corresponding timeSeriesData SHALL be writeable.
2: O	measurementId	<m*#{1..1}>	FOREIGN IDENTIFIER. If measurementId is used, it SHALL link to a measurement value of the total active power.
2: M	unit	"W"	The unit SHALL be applied to the value in timeSeriesData
2: M	scopeType	"activePowerLimit"	
3: M	TimeSeries. timeSeriesDescriptionListData. timeSeriesDescriptionData.		
3: M	timeSeriesId	<ts3#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
3: M	timeSeriesType	"fallbackConsumptionLimitCurve"	"fallbackConsumptionLimitCurve" SHALL be used to model the fallback consumption limit curve.
3: M	timeSeriesWriteable	"true"	If set to "true", the corresponding timeSeriesData SHALL be writeable.
3: O	measurementId	<m*#{1..1}>	FOREIGN IDENTIFIER. If measurementId is used, it SHALL link to a measurement value of the total active power.
3: M	unit	"W"	The unit SHALL be applied to the value in timeSeriesData
3: M	scopeType	"activePowerLimit"	
3: M	TimeSeries. timeSeriesDescriptionListData. timeSeriesDescriptionData.		
3: M	timeSeriesId	<ts4#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
3: M	timeSeriesType	"fallbackProductionLimitCurve"	"fallbackProductionLimitCurve" SHALL be used to model

			the fallback peoduction limit curve.
3: M	timeSeriesWriteable	"true"	If set to "true", the corresponding timeSeriesData SHALL be writeable.
3: O	measurementId	<m*#{1..1}>	FOREIGN IDENTIFIER. If measurementId is used, it SHALL link to a measurement value of the total active power.
3: M	unit	"W"	The unit SHALL be applied to the value in timeSeriesData
3: M	scopeType	"activePowerLimit"	

Table 17: Content of Function "timeSeriesDescriptionListData" at Actor CEM

3.2.2.2.2 Function "timeSeriesConstraintsListData"

Scenario {...}: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	TimeSeries. timeSeriesConstraintsListData. timeSeriesConstraintsData.		
1: M	timeSeriesId	<ts1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
1: O	slotCountMin		[POEN-013/1] If set, the corresponding timeSeriesData SHALL NOT have less slots.
1: M	slotCountMax	≥192	[POEN-013/2] If set, the corresponding timeSeriesData SHALL NOT have more slots.
1: O	slotDurationMin		[POEN-001/3], [POEN-013/3] If set, the corresponding timeSeriesData SHALL NOT have slots with shorter duration.
1: O	slotDurationMax		[POEN-001/3], [POEN-013/4] If set, the corresponding timeSeriesData SHALL NOT have slots with longer duration.
1: O	slotDurationStepSize		[POEN-013/5] If set, the duration of a slot SHALL be a multiple of slotDurationStepSize.
1: M	slotValueMin.* ¹		[POEN-002/1], [POEN-013/6] [Scaled number rules] The Elements "value", "maxValue" and "minValue" of the corresponding timeSeriesData's slots SHALL be equal to or higher than slotValueMin.

1: M	slotValueMin. number	0	SHALL be used.
1: O	slotValueMin. scale		MAY be used. If absent, a default value of "0" applies.
1: M	slotValueMax.* ¹		[POEN-002/1], [POEN-013/7] [Scaled number rules] The Elements "value", "maxValue" and "minValue" of the corresponding timeSeriesData's slots SHALL be equal to or lower than slotValueMax.
1: M	slotValueMax. number	≥0	SHALL be used.
1: O	slotValueMax. scale		MAY be used. If absent, a default value of "0" applies.
1: O	slotValueStepSize.		[POEN-002/2], [POEN-013/8] [Scaled number rules] The Elements "value", "maxValue" and "minValue" of the corresponding timeSeriesData's slots SHALL be a multiple of slotValueStepSize.
1: M	slotValueStepSize. number	≥0	SHALL be used.
1: O	slotValueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
2: M	TimeSeries. timeSeriesConstraintsListData. timeSeriesConstraintsData.		
2: M	timeSeriesId	<ts2#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: O	slotCountMin		[POEN-023/1] If set, the corresponding timeSeriesData SHALL NOT have less slots.
2: M	slotCountMax	≥192	[POEN-023/2] If set, the corresponding timeSeriesData SHALL NOT have more slots.
2: O	slotDurationMin		[POEN-001/3], [POEN-023/3] If set, the corresponding timeSeriesData SHALL NOT have slots with shorter duration.
2: O	slotDurationMax		[POEN-001/3], [POEN-023/4] If set, the corresponding timeSeriesData SHALL NOT have slots with longer duration.
2: O	slotDurationStepSize		[POEN-023/5] If set, the duration of a slot SHALL be a multiple of slotDurationStepSize.
2: M	slotValueMin.* ¹		[POEN-002/1], [POEN-023/6] [Scaled number rules] The Elements "value", "maxValue" and "minValue" of the corresponding timeSeriesData's slots SHALL be equal to or higher than slotValueMin.

2: M	slotValueMin. number	≤0	SHALL be used.
2: O	slotValueMin. scale		MAY be used. If absent, a default value of "0" applies.
2: M	slotValueMax.* ¹		[POEN-002/1], [POEN-023/7] [Scaled number rules] The Elements "value", "maxValue" and "minValue" of the corresponding timeSeriesData's slots SHALL be equal to or lower than slotValueMax.
2: M	slotValueMax. number	0	SHALL be used.
2: O	slotValueMax. scale		MAY be used. If absent, a default value of "0" applies.
2: O	slotValueStepSize.		[POEN-002/2], [POEN-023/8] [Scaled number rules] The Elements "value", "maxValue" and "minValue" of the corresponding timeSeriesData's slots SHALL be a multiple of slotValueStepSize.
2: M	slotValueStepSize. number	≥0	SHALL be used.
2: O	slotValueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
3: M	TimeSeries. timeSeriesConstraintsListData. timeSeriesConstraintsData.		
3: M	timeSeriesId	<ts3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: O	slotCountMin		[POEN-033/1] If set, the corresponding timeSeriesData SHALL NOT have less slots.
3: M	slotCountMax	≥96	[POEN-033/2] If set, the corresponding timeSeriesData SHALL NOT have more slots.
3: O	slotDurationMin		[POEN-001/3], [POEN-033/3] If set, the corresponding timeSeriesData SHALL NOT have slots with shorter duration.
3: O	slotDurationMax		[POEN-001/3], [POEN-033/4] If set, the corresponding timeSeriesData SHALL NOT have slots with longer duration.
3: O	slotDurationStepSize		[POEN-033/5] If set, the duration of a slot SHALL be a multiple of slotDurationStepSize.
3: M	slotValueMin.* ¹		[POEN-002/1], [POEN-033/6] [Scaled number rules] The Elements "value", "maxValue" and "minValue" of the corresponding timeSeriesData's slots SHALL be equal to or higher than slotValueMin.

3: M	slotValueMin. number	0	SHALL be used.
3: O	slotValueMin. scale		MAY be used. If absent, a default value of "0" applies.
3: M	slotValueMax.* ¹		[POEN-002/1], [POEN-033/7] [Scaled number rules] The Elements "value", "maxValue" and "minValue" of the corresponding timeSeriesData's slots SHALL be equal to or lower than slotValueMax.
3: M	slotValueMax. number	≥0	SHALL be used.
3: O	slotValueMax. scale		MAY be used. If absent, a default value of "0" applies.
3: O	slotValueStepSize.		[POEN-002/2], [POEN-033/8] [Scaled number rules] The Elements "value", "maxValue" and "minValue" of the corresponding timeSeriesData's slots SHALL be a multiple of slotValueStepSize.
3: M	slotValueStepSize. number	≥0	SHALL be used.
3: O	slotValueStepSize. scale		MAY be used. If absent, a default value of "0" applies.
3: M	TimeSeries. timeSeriesConstraintsListData. timeSeriesConstraintsData.		
3: M	timeSeriesId	<ts4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: O	slotCountMin		[POEN-034/1] If set, the corresponding timeSeriesData SHALL NOT have less slots.
3: M	slotCountMax	≥96	[POEN-034/2] If set, the corresponding timeSeriesData SHALL NOT have more slots.
3: O	slotDurationMin		[POEN-001/3], [POEN-034/3] If set, the corresponding timeSeriesData SHALL NOT have slots with shorter duration.
3: O	slotDurationMax		[POEN-001/3], [POEN-034/4] If set, the corresponding timeSeriesData SHALL NOT have slots with longer duration.
3: O	slotDurationStepSize		[POEN-034/5] If set, the duration of a slot SHALL be a multiple of slotDurationStepSize.
3: M	slotValueMin.* ¹		[POEN-002/1], [POEN-034/6] [Scaled number rules] The Elements "value", "maxValue" and "minValue" of the corresponding timeSeriesData's slots SHALL be equal to or higher than slotValueMin.

3: M	slotValueMin. number	≤0	SHALL be used.
3: O	slotValueMin. scale		MAY be used. If absent, a default value of "0" applies.
3: M	slotValueMax.* ¹		[POEN-002/1], [POEN-034/7] [Scaled number rules] The Elements "value", "maxValue" and "minValue" of the corresponding timeSeriesData's slots SHALL be equal to or lower than slotValueMax.
3: M	slotValueMax. number	0	SHALL be used.
3: O	slotValueMax. scale		MAY be used. If absent, a default value of "0" applies.
3: O	slotValueStepSize.		[POEN-002/2], [POEN-034/8] [Scaled number rules] The Elements "value", "maxValue" and "minValue" of the corresponding timeSeriesData's slots SHALL be a multiple of slotValueStepSize.
3: M	slotValueStepSize. number	≥0	SHALL be used.
3: O	slotValueStepSize. scale		MAY be used. If absent, a default value of "0" applies.

Table 18: Content of Function "timeSeriesConstraintsListData" at Actor CEM

*¹: The Elements slotValueMin and slotValueMax SHALL be interpreted mathematically.

3.2.2.2.3 Function "timeSeriesListData"

Scenario [{...}]: M/R/O [W][C]	Element	Value	[High-Level Mapping] Element and value rules
1: M	TimeSeries. timeSeriesListData. timeSeriesData.		
1: M	timeSeriesId	<ts1#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
1: F	timePeriod.		
1: M \RW	timeSeriesSlot.		
1: M \RW	timeSeriesSlot. timeSeriesSlotId		
1: M \RW	timeSeriesSlot. timePeriod.		
1: M \RW	timeSeriesSlot. timePeriod. startTime	<xs:dateTime>	[POEN-001/1]
1: M \RW	timeSeriesSlot. duration	>0	[POEN-001/3], [POEN-005/1], [POEN-005/2] SHALL be set and express the duration of the slot. SHALL only contain values

			greater than zero seconds. The duration SHALL last to the start time of the next slot, if any.
1: M \RW	timeSeriesSlot. minValue.* ¹		[POEN-002/1], [POEN-006], [POEN-011] [Scaled number rules]
1: M \RW	timeSeriesSlot. minValue. number	>0	SHALL be used.
1: M \RW	timeSeriesSlot. minValue. scale		SHALL be used.
1: M \RW	timeSeriesSlot. maxValue.* ¹		[POEN-002/1], [POEN-006], [POEN-012] [Scaled number rules]
1: M \RW	timeSeriesSlot. maxValue. number	>0	SHALL be used.
1: M \RW	timeSeriesSlot. maxValue. scale		SHALL be used.
2: M	TimeSeries. timeSeriesListData. timeSeriesData.		
2: M	timeSeriesId	<ts2#{1..1}>	SHALL be set as PRIMARY IDENTIFIER.
2: F	timePeriod.		
2: M \RW	timeSeriesSlot.		
2: M \RW	timeSeriesSlot. timeSeriesSlotId		
2: M \RW	timeSeriesSlot. timePeriod.		
2: M \RW	timeSeriesSlot. timePeriod. startTime	<xs:dateTime>	[POEN-001/1]
2: M \RW	timeSeriesSlot. duration	>0	[POEN-001/3], [POEN-005/1], [POEN-005/2] SHALL be set and express the duration of the slot. SHALL only contain values greater than zero seconds. The duration SHALL last to the start time of the next slot, if any.
2: M \RW	timeSeriesSlot. minValue.* ²		[POEN-002/1], [POEN-006], [POEN-021] [Scaled number rules]
2: M \RW	timeSeriesSlot. minValue. number	≤0	SHALL be used.
2: M \RW	timeSeriesSlot. minValue. scale		SHALL be used.
2: M \RW	timeSeriesSlot. maxValue.* ²		[POEN-002/1], [POEN-006], [POEN-022] [Scaled number rules]
2: M \RW	timeSeriesSlot. maxValue. number	≤0	SHALL be used.
2: M \RW	timeSeriesSlot. maxValue. scale		SHALL be used.

3: M	TimeSeries. timeSeriesListData. timeSeriesData.		
3: M	timeSeriesId	<ts3#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: F	timePeriod.		
3: M \RW	timeSeriesSlot.		[POEN-009]
3: M \RW	timeSeriesSlot. timeSeriesSlotId		
3: M \RW	timeSeriesSlot. timePeriod.		
3: M \RW	timeSeriesSlot. timePeriod. startTime	<xs:time>	[POEN-001/1], [POEN-001/2]
3: M \RW	timeSeriesSlot. duration	>0	[POEN-001/3], [POEN-005/1], [POEN-005/2] SHALL be set and express the duration of the slot. SHALL only contain values greater than zero seconds. The duration SHALL last to the start time of the next slot, if any.
3: M \RW	timeSeriesSlot. maxValue.* ³		[POEN-002/1], [POEN-006], [POEN-007], [POEN-008], [POEN-031] [Scaled number rules]
3: M \RW	timeSeriesSlot. maxValue. number	≥0	SHALL be used.
3: M \RW	timeSeriesSlot. maxValue. scale		SHALL be used.
3: M	TimeSeries. timeSeriesListData. timeSeriesData.		
3: M	timeSeriesId	<ts4#(1..1)>	SHALL be set as PRIMARY IDENTIFIER.
3: F	timePeriod.		
3: M \RW	timeSeriesSlot.		[POEN-009]
3: M \RW	timeSeriesSlot. timeSeriesSlotId		
3: M \RW	timeSeriesSlot. timePeriod.		
3: M \RW	timeSeriesSlot. timePeriod. startTime	<xs:time>	[POEN-001/1], [POEN-001/2]
3: M \RW	timeSeriesSlot. duration	>0	[POEN-001/3], [POEN-005/1], [POEN-005/2] SHALL be set and express the duration of the slot. SHALL only contain values greater than zero seconds. The duration SHALL last to the start time of the next slot, if any.
3: M \RW	timeSeriesSlot. maxValue.* ⁴		[POEN-002/1], [POEN-006], [POEN-007], [POEN-008], [POEN-032] [Scaled number rules]
3: M \RW	timeSeriesSlot. maxValue. number	≤0	SHALL be used.
3: M \RW	timeSeriesSlot. maxValue. scale		SHALL be used.

Table 19: Content of Function "timeSeriesListData" at Actor CEM

*1: The Elements minVal and maxVal SHALL be interpreted logically (which corresponds to the mathematical way for the consumption direction).

1387 *²: The Elements min**Value** and max**Value** SHALL be interpreted logically (not mathematically). This
1388 means that the min**Value** is mathematically higher than the max**Value**.

1389 *³: The Element max**Value** SHALL be interpreted logically (which corresponds to the mathematical
1390 way for the consumption direction).

1391 *⁴: The Element max**Value** SHALL be interpreted logically (not mathematically). This means that the
1392 max**Value** is the mathematically smallest value that is permitted for the power production (as
1393 production power is provided as negative values).

1394

1395 **3.2.2.3 Client data - Specializations**

1396 As this Actor has only server functionality, no Specializations are described within this section.

1397

1398 **3.3 Pre-Scenario communication**

1399 **3.3.1 General information**

1400 The Pre-Scenario communication is needed if a client does not know the corresponding addresses on
1401 the server or if the required subscriptions or bindings are not active. In this case certain general
1402 communication mechanisms SHALL be used within SPINE:

- 1403 a) Detailed discovery: allows to discover resource addresses.
- 1404 b) Binding: allows to bind to resource address, which is frequently necessary to obtain write
1405 privileges.
- 1406 c) Subscription: allows to subscribe to resource addresses, which is necessary to receive
1407 unsolicited notifications if a resource changes during runtime.

1408 It is possible to combine those steps for multiple Scenarios or also multiple Use Cases:

- 1409 - E.g. if multiple Scenarios in multiple Use Cases use the same Feature, only one subscription
1410 needs to occur.
- 1411 - E.g. a complete detailed discovery or a subscription to the NodeManagement Feature needs
1412 to occur only once for all Use Cases.

1413 Depending on which Entity, Feature and Functions are used within a Scenario the payload of the
1414 corresponding messages may slightly differ, but the basic principles and messages used stay the
1415 same.

1416 The subsequent messages SHALL be exchanged for those parts that have not already been performed
1417 since the current connection is established or if those parts are outdated for another reason (e.g. if
1418 the detailed discovery is needed, but the bindings and subscriptions are still active from a previous
1419 connection only the detailed discovery messages need to be exchanged). If all Pre-Scenario
1420 communication parts are up to date, this section MAY be skipped, and the implementation can
1421 proceed as described in the corresponding "Scenario communication" sections.

1422 After the connection is re-established (e.g. due to a power loss or a firmware update) the Pre-
1423 Scenario communication SHALL be performed as well. There may be circumstances where messages
1424 from the Pre-Scenario communication may be exchanged again.

1425 Often the necessary messages of different Scenarios can be combined, so that only one single
1426 message is needed instead of multiple messages for the different Scenarios. This also is the case for
1427 the Pre-Scenario communication. In most cases only one "read" operation on the detailed discovery
1428 is necessary, as well as only one subscription request or binding request is needed for each Feature.
1429 Often multiple Scenarios within a Use Case access the same Feature, so only one subscription or
1430 binding is necessary.

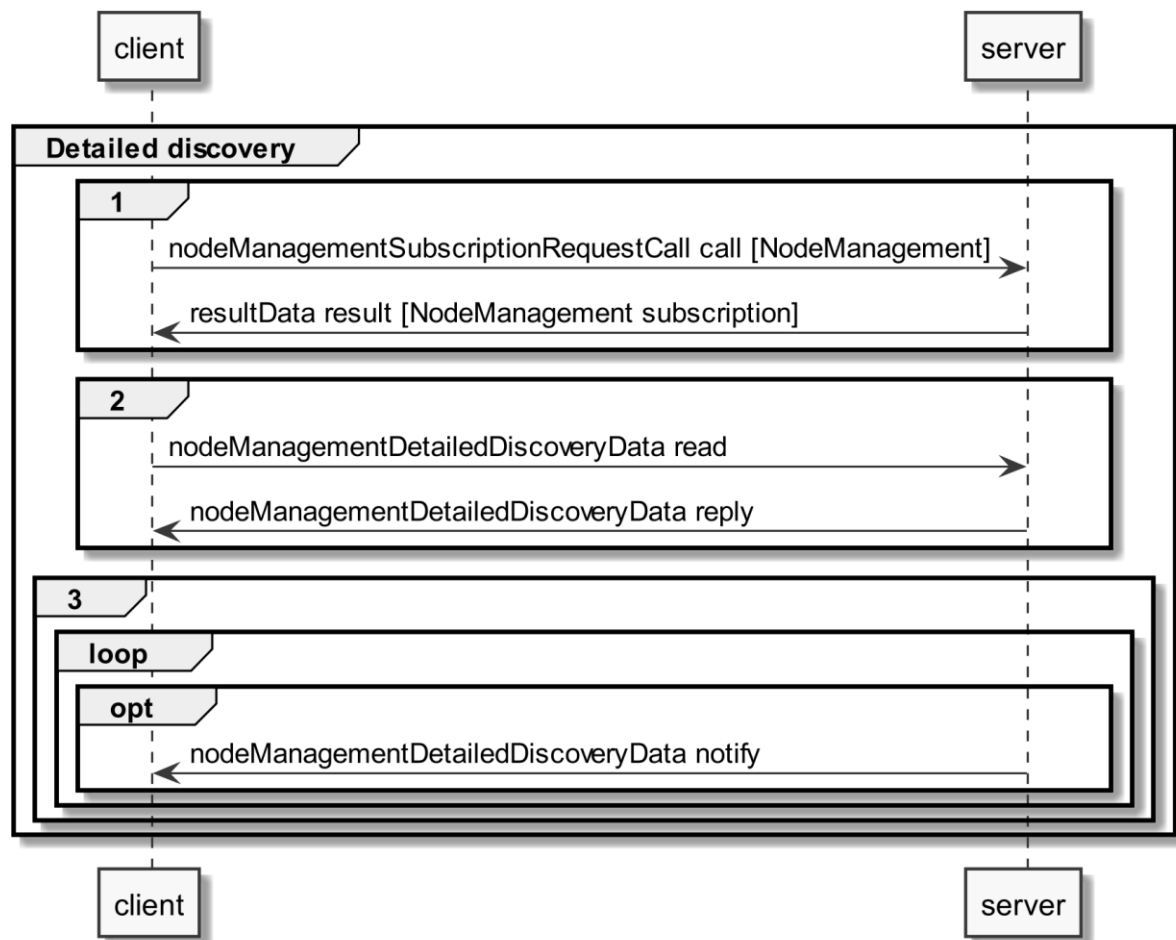
1431

1432 **3.3.2 Detailed discovery**

1433 For the functionality where a client already has current detailed discovery information (i.e.
1434 independent of this Use Case or any Scenario of it) the remainder of this section SHOULD be skipped.

1435 Otherwise, the following procedure SHALL be performed in the given order:

- 1436 1. If a client is not subscribed to the primary NodeManagement instance, the client SHALL
1437 acquire a subscription according to the figure provided within this sub-section.
- 1438 2. A client SHALL read the detailed discovery information according to the figure provided
1439 within this sub-section. It SHALL keep the received information as far as it concerns
1440 mandatory and supported optional Entity Types, Feature Types and Functions of this Use
1441 Case that are needed by the client. This means that a client may choose how to store the
1442 necessary information. E.g. a client Actor can store the information how to address the
1443 necessary Features of the implemented Scenarios but may discard the Entity information.
- 1444 3. If and as long as a client has a subscription to the detailed discovery information of an Actor
1445 and receives proper notifications, it SHALL consider (i.e. integrate into the kept detailed
1446 discovery information) the received information as far as it concerns mandatory and
1447 supported optional Entity Types, Feature Types and Functions of this Use Case.



1448

1449 *Figure 10: Pre-Scenario communication - Detailed discovery sequence diagram*

1450 If the "nodeManagementDetailedDiscoveryData read" fails, the client SHOULD retry to read the
 1451 detailed discovery information until the "nodeManagementDetailedDiscoveryData reply" message
 1452 was received successfully.

1453 If all functionality is present at all times: The "nodeManagementDetailedDiscoveryData reply"
 1454 message contains at least the mandatory Entities and Features given in the "Actor [...] overview"
 1455 diagrams as well as the used Functions and their "possible operations" described in section 3.2 and
 1456 its sub-sections.

1457 If functionality is added or removed dynamically: The "nodeManagementDetailedDiscoveryData
 1458 reply" message does not need to contain all mandatory Entities and Features given in the "Actor [...]
 1459 overview" diagrams as well as all needed Functions and their "possible operations" described in
 1460 section 3.2 and its sub-sections. However, as soon as the functionality is available it will be
 1461 announced via a "nodeManagementDetailedDiscoveryData notify" message.

1462 For the nodeManagementDetailedDiscoveryData read Function it is recommended to use a partial
 1463 read with separated Selectors that may use one of the following Elements:

- 1464 - entityType
- 1465 - featureType

1466 Note: Even with the usage of Selectors Features and Entities that are not relevant for this Use Case
 1467 may be discovered. However, only Features and Entities that fulfil the hierarchical order as described
 1468 within the Actors' sections shall be considered for this Use Case.

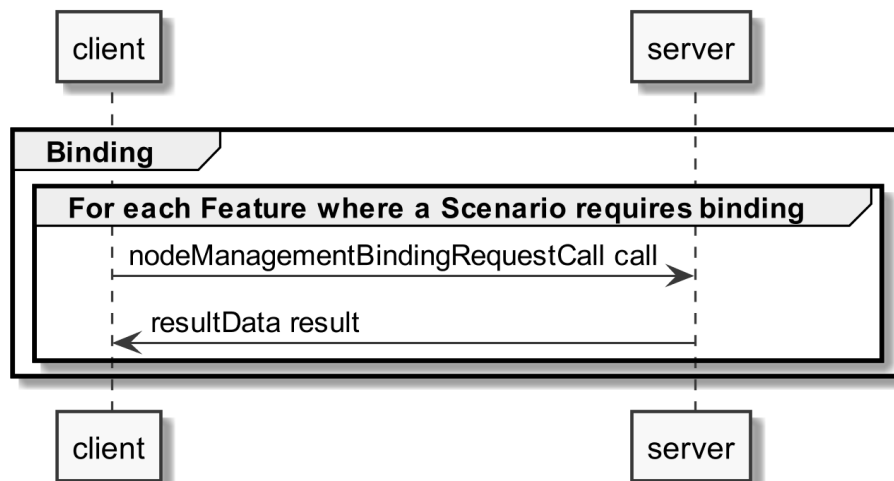
1469 A "partial" notify SHALL be supported without using Selectors and Elements. Partial "delete" notify
 1470 SHOULD also be supported with separated Selectors that may use one of the following Elements:

- 1471 - entityAddress
- 1472 - featureAddress

1473

1474 3.3.3 Binding

1475 If binding is required by a Scenario that uses Features with writeable or changeable data, the server
 1476 SHALL support binding for the respective Features. Before a write on a Function of a Feature occurs,
 1477 the client SHALL create a binding to the corresponding Feature. For this the
 1478 nodeManagementBindingRequestCall Function is used as shown in the following sequence diagram:



1479

1480 *Figure 11: Pre-Scenario communication - Binding sequence diagram*

1481 If functionality is added or removed dynamically, binding may not be possible at all times on the
 1482 required Functions. A client SHALL retry to create a binding again when receiving according updated
 1483 detailed discovery information.

1484

1485 3.3.4 Subscription

1486 A server SHALL support subscription for all Features that contain readable data that may change
 1487 during runtime. The client SHALL create a subscription for all Features that the client wants to read.
 1488 For this the nodeManagementSubscriptionRequestCall Function is used as shown in the following
 1489 sequence diagram:

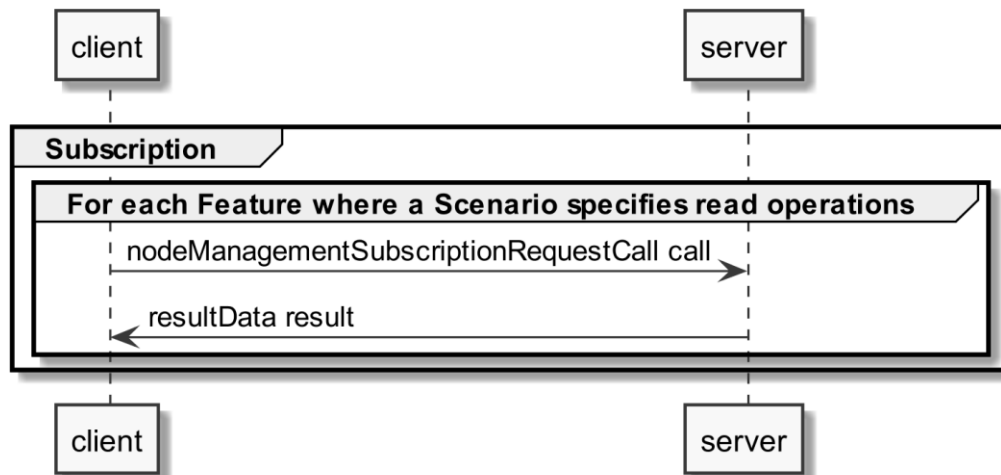


Figure 12: Pre-Scenario communication - Subscription sequence diagram

If the subscription request fails (e.g. because it is not supported by the server or the maximum number of possible subscriptions is reached), the client SHOULD read the data periodically (so-called "polling").

If functionality is added or removed dynamically, subscription may not be possible at all times on the required Functions. A client SHALL retry its subscription procedure again when receiving according updated detailed discovery information.

3.3.5 Dynamic behaviour

In case Entities or Features are removed, a nodeManagementDetailedDiscoveryData "notify" is transmitted that informs about the deleted Entities and Features. All existing binding or subscription entries on the deleted Features SHALL be deleted by each device.

In case Entities or Features are added the Pre-Scenario communication starts with transmitting a nodeManagementDetailedDiscoveryData "notify" that contains the added Entities and Features.

3.4 Scenarios

3.4.1 Scenario 1 - Set active power consumption limit curves

3.4.1.1 Pre-Scenario communication

1. **Detailed discovery:** Actors that act as client within this Scenario, need to know the addresses of the server Features used in the Initial Scenario communication. If the address of a particular server Feature is not known, the detailed discovery must be used, as described in section 3.3.2.
2. **Binding:** Actors that write parts of a Feature within this Scenario, need to create a binding, as described in section 3.3.3. Only one binding partner is allowed to write the data specified in this Scenario.
3. **Subscription:** Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as soon as the address of a required resource is known, the Initial Scenario communication for this resource MAY start already, even if the addresses of other required resources are not known yet.

If required resources are removed and added again, they are re-discovered, and the Initial Scenario communication is triggered again for those resources.

3.4.1.2 Initial Scenario communication

Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped, the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding resources may have changed in the meantime:

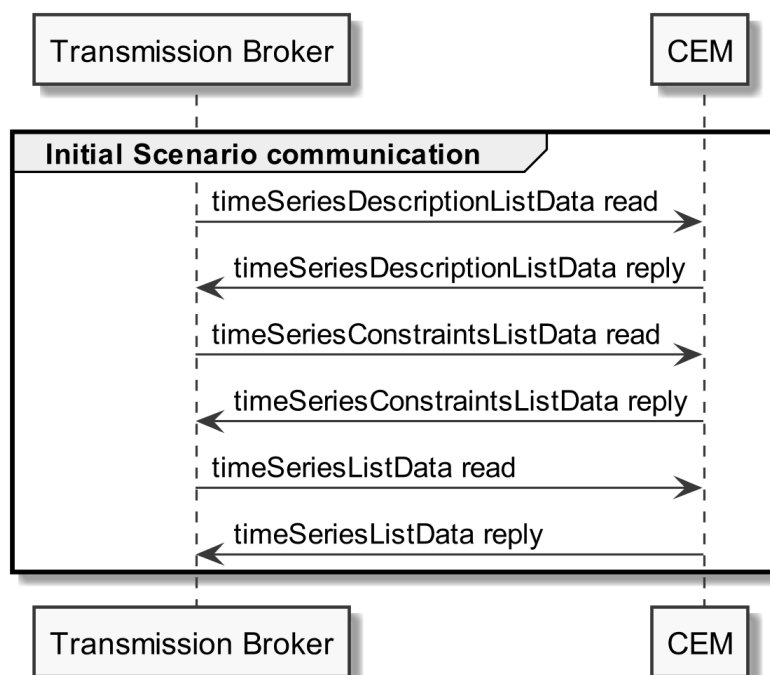


Figure 13: Scenario 1 - Initial Scenario communication sequence diagram

Partial data information regarding [POEN-011], [POEN-012] and [POEN-013]:

The `timeSeriesDescriptionListData read` SHOULD be a "partial" read operation with the following Selector:

- `timeSeriesType = "consumptionLimitCurve"`
- `scopeType = "activePowerLimit"`

The `timeSeriesConstraintsListData read` and `timeSeriesListData read` SHOULD be "partial" read operation with the following Selector:

- `timeSeriesId` (derived from the `timeSeriesDescriptionListData reply`)

Note: If partial read is not supported, a full read SHALL be performed.

1541 The following table shows where the required content of the messages from the sequence diagram is
 1542 described:

Message name from sequence diagram	Content description in table	Scenario number in table
timeSeriesDescriptionListData reply	Table 17	1
timeSeriesConstraintsListData reply	Table 18	1
timeSeriesListData reply	Table 19	1

1543 *Table 20: Initial Scenario communication content references for Scenario 1*

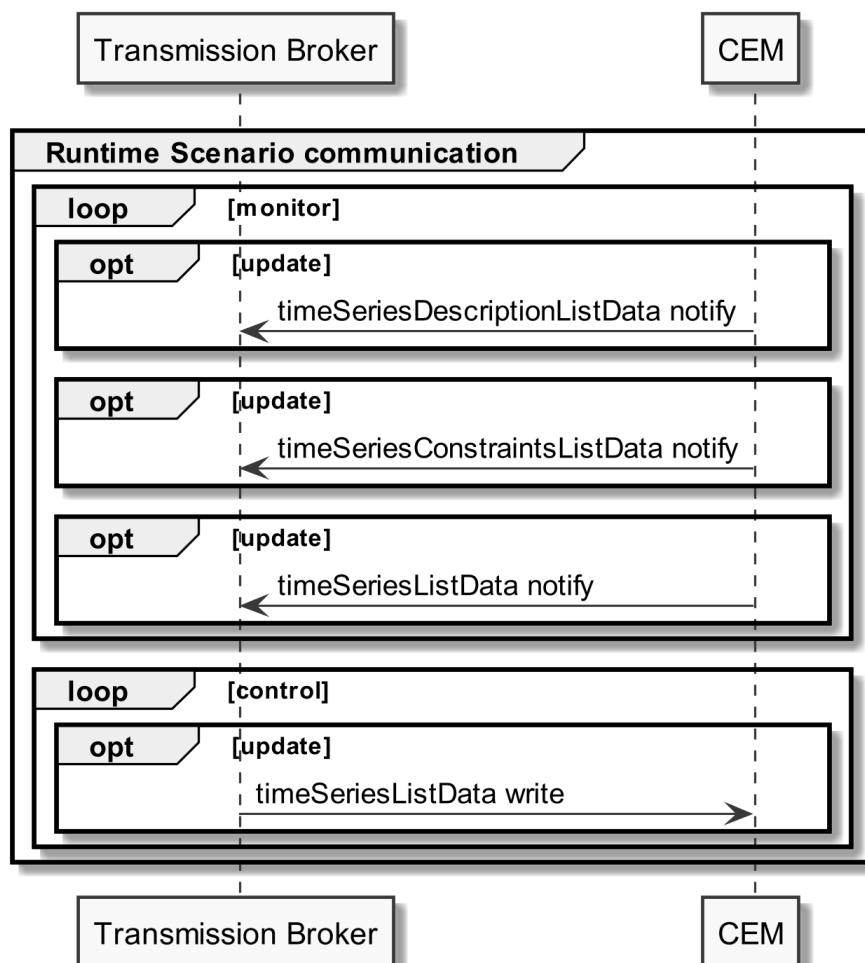
1544 Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be
 1545 provided completely, but later during Runtime Scenario communication.

1546

1547 **3.4.1.3 Runtime Scenario communication**

1548 Based on the Initial Scenario communication, the Runtime Scenario communication provides updates
 1549 during runtime.

1550 If one of the referenced server Functions' data change, the server SHALL submit the change as shown
 1551 in the following figure:



1552

1553 *Figure 14: Scenario 1 - Runtime Scenario communication sequence diagram*

1554 Note: Normally, in this Scenario only the "timeSeriesListData" Function changes during runtime.
 1555 Hence, usually no notifications of the other Functions of this Scenario are sent during runtime.

1556 Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this
 1557 Scenario.

1558 For timeSeriesDescriptionListData notify, timeSeriesConstraintsListData notify and timeSeriesListData
 1559 notify "partial" delete notifications SHOULD be supported with the Selector:

1560 - timeSeriesId

1561 Note: To interpret partial notification messages correctly, the information obtained during the Initial
 1562 Scenario communication phase is required.

1563 Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could
 1564 not be evaluated.

1565

1566 The following table shows where the required content of the messages of the sequence diagram is
 1567 described:

Message name from sequence diagram	Content description in table	Scenario number in table
timeSeriesDescriptionListData reply	Table 17	1
timeSeriesConstraintsListData reply	Table 18	1
timeSeriesListData reply	Table 19	1
timeSeriesListData write	Table 19	1

1568 *Table 21: Runtime Scenario communication content references for Scenario 1*

1569

1570 **3.4.1.4 Additional information**

1571 If upon receipt the limit curve cannot be applied by the CEM, the CEM SHALL respond with an
 1572 "Application error" message (NACK, [POEN-004/1]) using the "result" classifier and resultData
 1573 containing the errorNumber 7 ("Command rejected") (see "Acknowledgement concept" in
 1574 [ProtocolSpecification]).

1575

1576 **3.4.2 Scenario 2 - Set active power production limit curves**

1577 **3.4.2.1 Pre-Scenario communication**

- 1578 1. **Detailed discovery:** Actors that act as client within this Scenario, need to know the addresses
 1579 of the server Features used in the Initial Scenario communication. If the address of a
 1580 particular server Feature is not known, the detailed discovery must be used, as described in
 1581 section 3.3.2.
- 1582 2. **Binding:** Actors that write parts of a Feature within this Scenario, need to create a binding, as
 1583 described in section 3.3.3. Only one binding partner is allowed to write the data specified in
 1584 this Scenario.

3. **Subscription:** Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as soon as the address of a required resource is known, the Initial Scenario communication for this resource MAY start already, even if the addresses of other required resources are not known yet.

If required resources are removed and added again, they are re-discovered, and the Initial Scenario communication is triggered again for those resources.

3.4.2.2 Initial Scenario communication

Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped, the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding resources may have changed in the meantime:

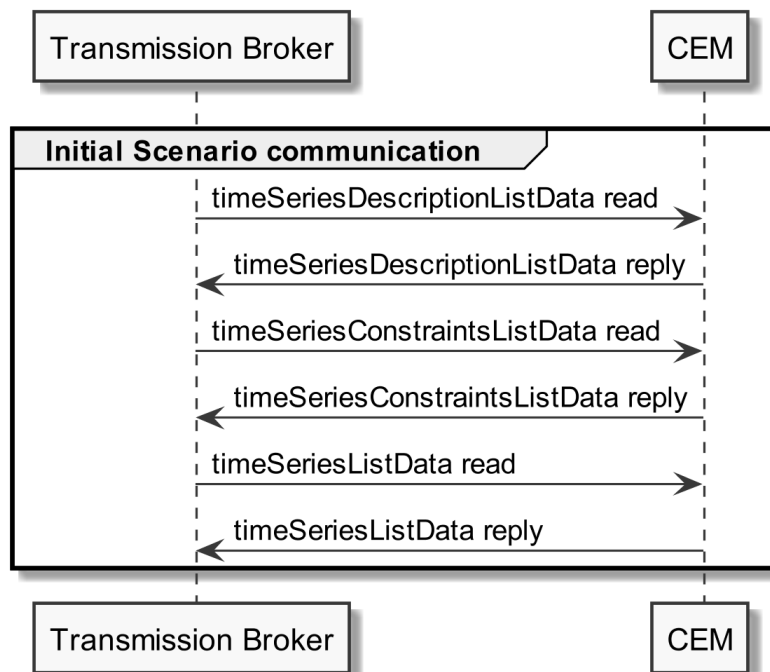


Figure 15: Scenario 2 - Initial Scenario communication sequence diagram

Partial data information regarding [POEN-021], [POEN-022] and [POEN-023]:

The `timeSeriesDescriptionListData read` SHOULD be a "partial" read operation with the following Selector:

- `timeSeriesType` = "productionLimitCurve"
- `scopeType` = "activePowerLimit"

The `timeSeriesConstraintsListData read` and `timeSeriesListData read` SHOULD be "partial" read operation with the following Selector:

- `timeSeriesId` (derived from the `timeSeriesDescriptionListData reply`)

Note: If partial read is not supported, a full read SHALL be performed.

The following table shows where the required content of the messages from the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
timeSeriesDescriptionListData reply	Table 17	2
timeSeriesConstraintsListData reply	Table 18	2
timeSeriesListData reply	Table 19	2

Table 22: Initial Scenario communication content references for Scenario 2

Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be provided completely, but later during Runtime Scenario communication.

3.4.2.3 Runtime Scenario communication

Based on the Initial Scenario communication, the Runtime Scenario communication provides updates during runtime.

If one of the referenced server Functions' data change, the server SHALL submit the change as shown in the following figure:

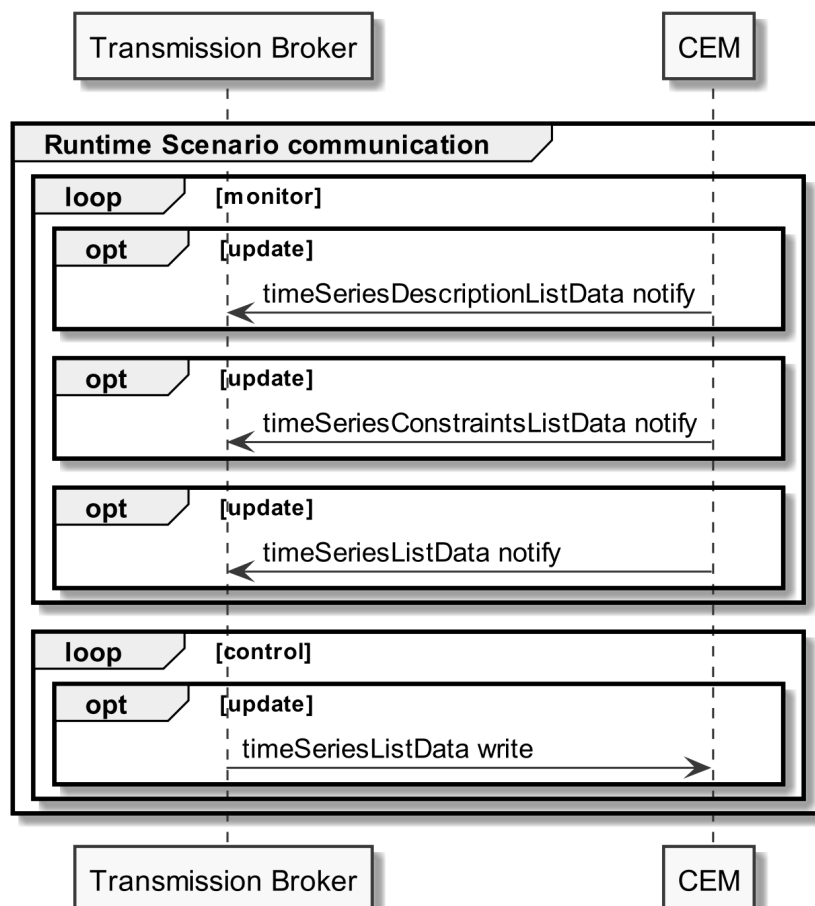


Figure 16: Scenario 2 - Runtime Scenario communication sequence diagram

1623 Note: Normally, in this Scenario only the "timeSeriesListData" Function changes during runtime.
 1624 Hence, usually no notifications of the other Functions of this Scenario are sent during runtime.

1625 Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this
 1626 Scenario.

1627 For timeSeriesDescriptionListData notify, timeSeriesConstraintsListData notify and timeSeriesListData
 1628 notify "partial" delete notifications SHOULD be supported with the Selector:

1629 - timeSeriesId

1630 Note: To interpret partial notification messages correctly, the information obtained during the Initial
 1631 Scenario communication phase is required.

1632 Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could
 1633 not be evaluated.

1634

1635 The following table shows where the required content of the messages of the sequence diagram is
 1636 described:

Message name from sequence diagram	Content description in table	Scenario number in table
timeSeriesDescriptionListData reply	Table 17	2
timeSeriesConstraintsListData reply	Table 18	2
timeSeriesListData reply	Table 19	2
timeSeriesListData write	Table 19	2

1637 *Table 23: Runtime Scenario communication content references for Scenario 2*

1638

1639 **3.4.2.4 Additional information**

1640 If upon receipt the limit curve cannot be applied by the CEM, the CEM SHALL respond with an
 1641 "Application error" message (NACK, [POEN-004/2]) using the "result" classifier and resultData
 1642 containing the errorNumber 7 ("Command rejected") (see "Acknowledgement concept" in
 1643 [ProtocolSpecification]).

1644

1645 **3.4.3 Scenario 3 - Fallback limit curves**

1646 **3.4.3.1 Pre-Scenario communication**

- 1647 1. **Detailed discovery:** Actors that act as client within this Scenario, need to know the addresses
 1648 of the server Features used in the Initial Scenario communication. If the address of a
 1649 particular server Feature is not known, the detailed discovery must be used, as described in
 1650 section 3.3.2.
- 1651 2. **Binding:** Actors that write parts of a Feature within this Scenario, need to create a binding, as
 1652 described in section 3.3.3. Only one binding partner is allowed to write the data specified in
 1653 this Scenario.

3. **Subscription:** Actors SHALL create a subscription for each server Feature that is relevant for the corresponding Actor within this Scenario, as described in section 3.3.4.

The Initial Scenario communication SHALL start at the latest when the required resources on an Actor are known and the necessary binding and subscription procedures have been finished. However, as soon as the address of a required resource is known, the Initial Scenario communication for this resource MAY start already, even if the addresses of other required resources are not known yet.

If required resources are removed and added again, they are re-discovered, and the Initial Scenario communication is triggered again for those resources.

3.4.3.2 Initial Scenario communication

Each time a (re-)connection is established, even if the Pre-Scenario communication phase is skipped, the messages shown in the following sequence diagram SHALL be exchanged, as the corresponding resources may have changed in the meantime:

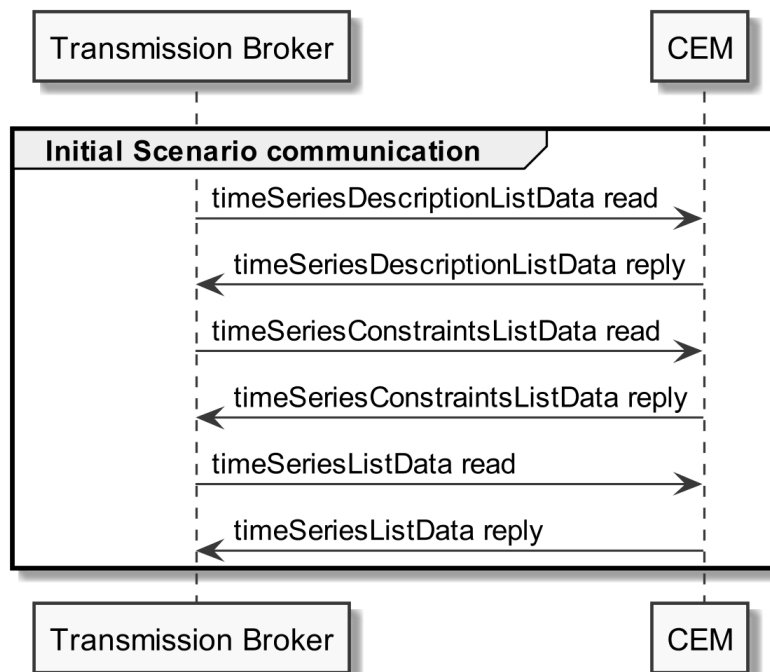


Figure 17: Scenario 3 - Initial Scenario communication sequence diagram

Partial data information regarding [POEN-031] and [POEN-033]:

The `timeSeriesDescriptionListData read` SHOULD be a "partial" read operation with the following Selector:

- `timeSeriesType` = "fallbackConsumptionLimitCurve"
- `scopeType` = "activePowerLimit"

The `timeSeriesConstraintsListData read` and `timeSeriesListData read` SHOULD be "partial" read operation with the following Selector:

- `timeSeriesId` (derived from the `timeSeriesDescriptionListData reply`)

1677 Note: If partial read is not supported, a full read SHALL be performed.

1678

1679 **Partial data information regarding [POEN-032] and [POEN-034]:**

1680 The timeSeriesDescriptionListData read SHOULD be a "partial" read operation with the following
1681 Selector:

- 1682 - timeSeriesType = "fallbackProductionLimitCurve"
- 1683 - scopeType = "activePowerLimit"

1684 The timeSeriesConstraintsListData read and timeSeriesListData read SHOULD be "partial" read
1685 operation with the following Selector:

- 1686 - timeSeriesId (derived from the timeSeriesDescriptionListData reply)

1687 Note: If partial read is not supported, a full read SHALL be performed.

1688 The following table shows where the required content of the messages from the sequence diagram is
1689 described:

Message name from sequence diagram	Content description in table	Scenario number in table
timeSeriesDescriptionListData reply	Table 17	3
timeSeriesConstraintsListData reply	Table 18	3
timeSeriesListData reply	Table 19	3

1690 *Table 24: Initial Scenario communication content references for Scenario 3*

1691 Note: Within the Initial Scenario communication, the content required by this Scenario MAY not be
1692 provided completely, but later during Runtime Scenario communication.

1693

1694 **3.4.3.3 Runtime Scenario communication**

1695 Based on the Initial Scenario communication, the Runtime Scenario communication provides updates
1696 during runtime.

1697 If one of the referenced server Functions' data change, the server SHALL submit the change as shown
1698 in the following figure:

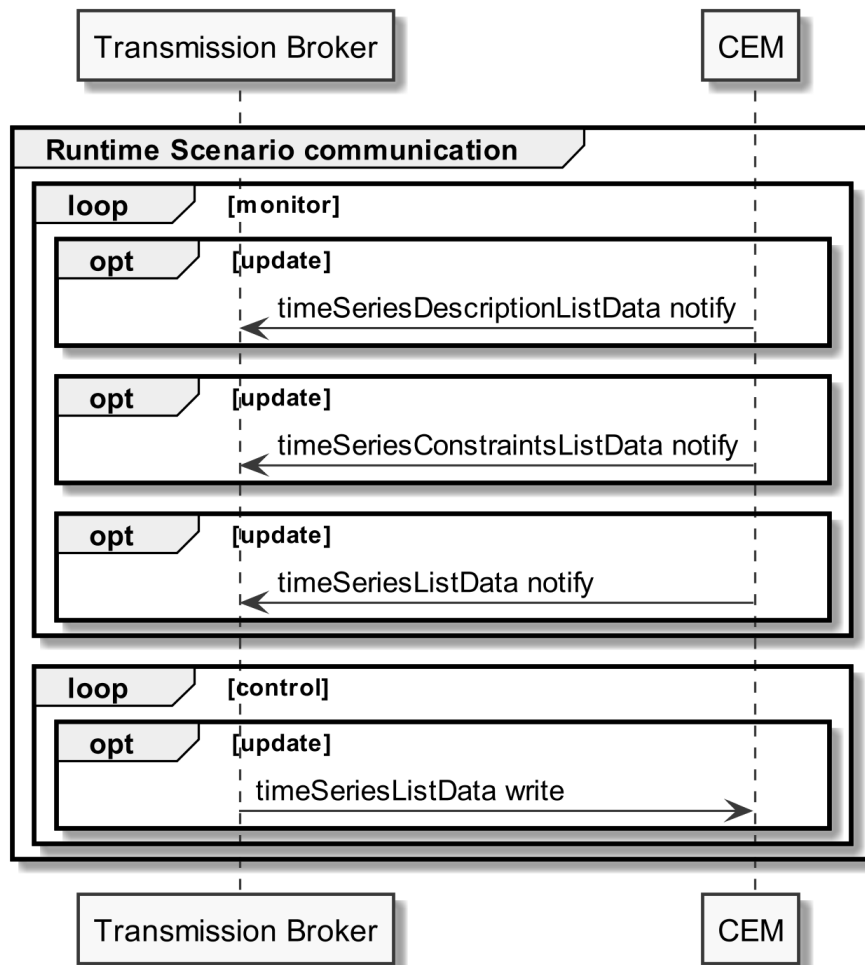


Figure 18: Scenario 3 - Runtime Scenario communication sequence diagram

Note: Normally, in this Scenario only the "timeSeriesListData" Function changes during runtime. Hence, usually no notifications of the other Functions of this Scenario are sent during runtime.

Partial notifications without Selectors or Elements SHALL be supported for all Functions used in this Scenario.

For timeSeriesDescriptionListData notify, timeSeriesConstraintsListData notify and timeSeriesListData notify "partial" delete notifications SHOULD be supported with the Selector:

- timeSeriesId

Note: To interpret partial notification messages correctly, the information obtained during the Initial Scenario communication phase is required.

Note: A read operation ("polling") on all Functions is possible at any time, e.g. if a notification could not be evaluated.

The following table shows where the required content of the messages of the sequence diagram is described:

Message name from sequence diagram	Content description in table	Scenario number in table
timeSeriesDescriptionListData reply	Table 17	3
timeSeriesConstraintsListData reply	Table 18	3
timeSeriesListData reply	Table 19	3
timeSeriesListData write	Table 19	3

Table 25: Runtime Scenario communication content references for Scenario 3

3.4.3.4 Additional information

If upon receipt the limit curve cannot be applied by the CEM, the CEM SHALL respond with an "Application error" message (NACK, [POEN-004/3]) using the "result" classifier and resultData containing the errorNumber 7 ("Command rejected") (see "Acknowledgement concept" in [ProtocolSpecification]).